

BFS Traversal

Step 1: Start at E

Legend

Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



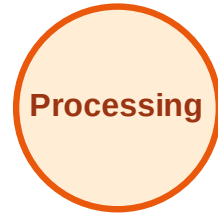
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



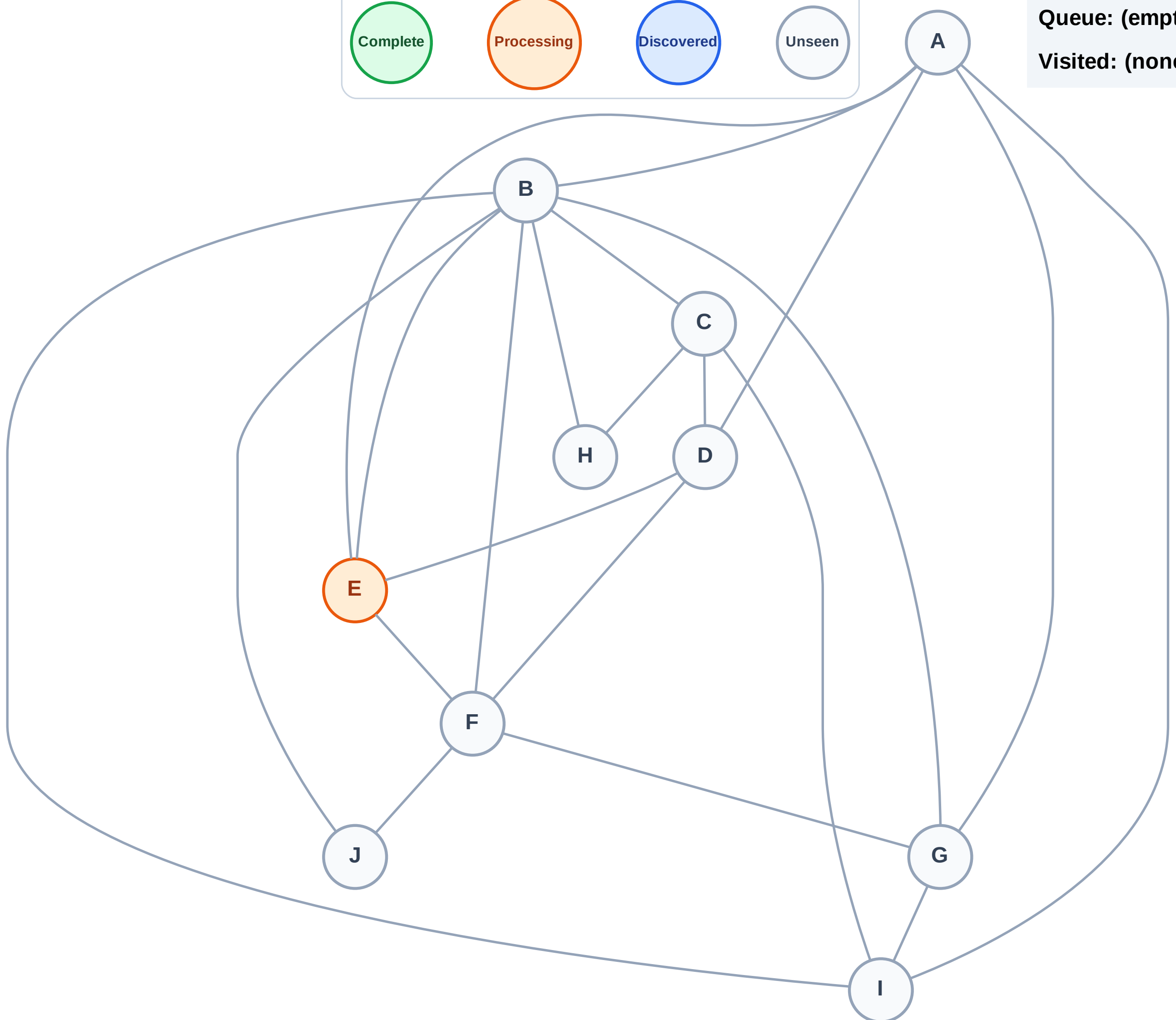
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

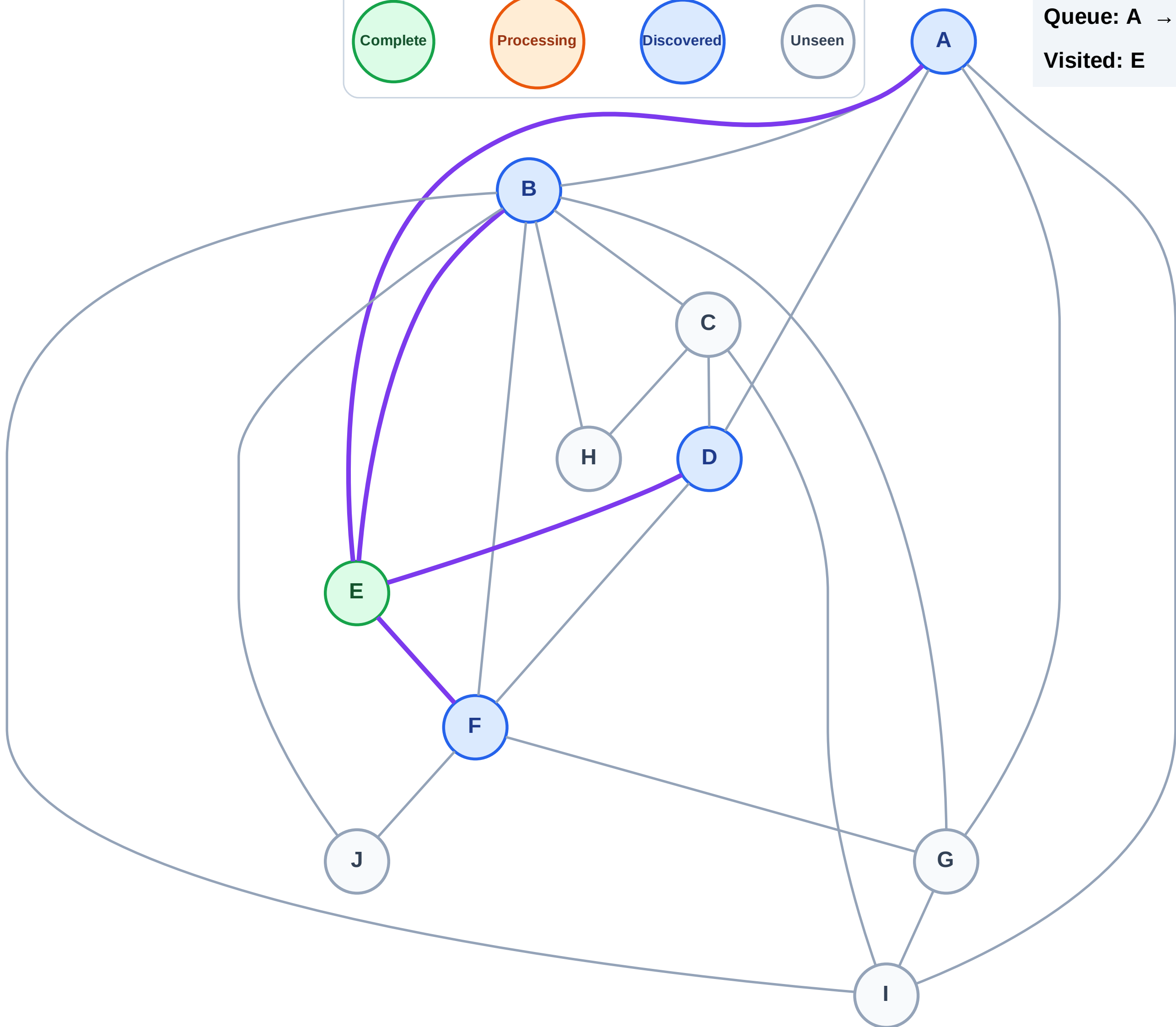
Step 3: Discover A, B, D, F from E

Legend



Queue: A → B → D → F

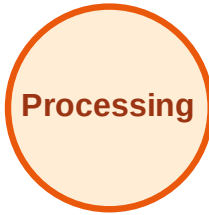
Visited: E



BFS Traversal

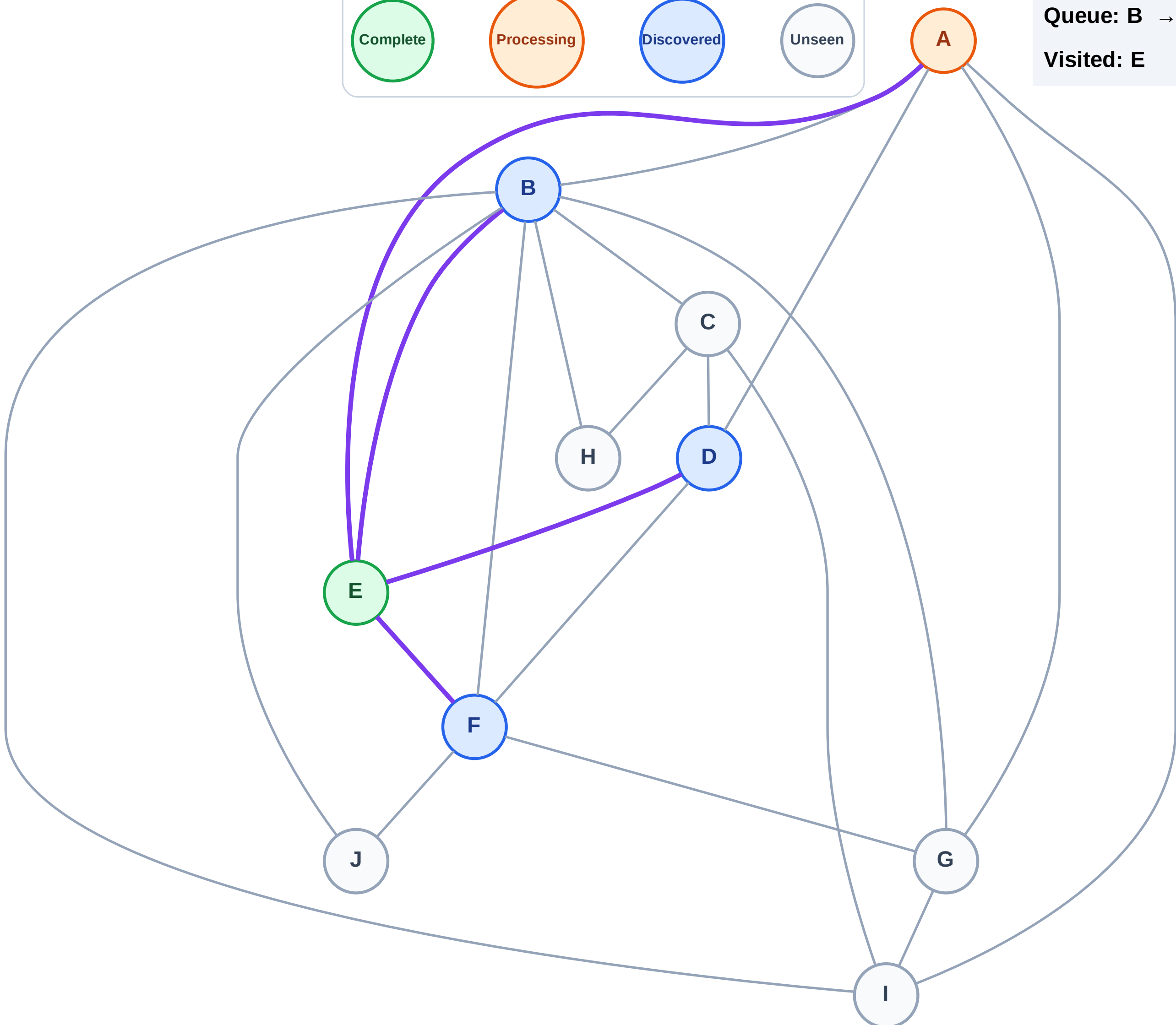
Step 4: Process node A

Legend



Queue: B → D → F

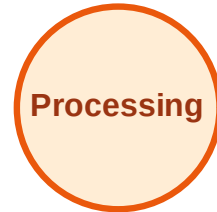
Visited: E



BFS Traversal

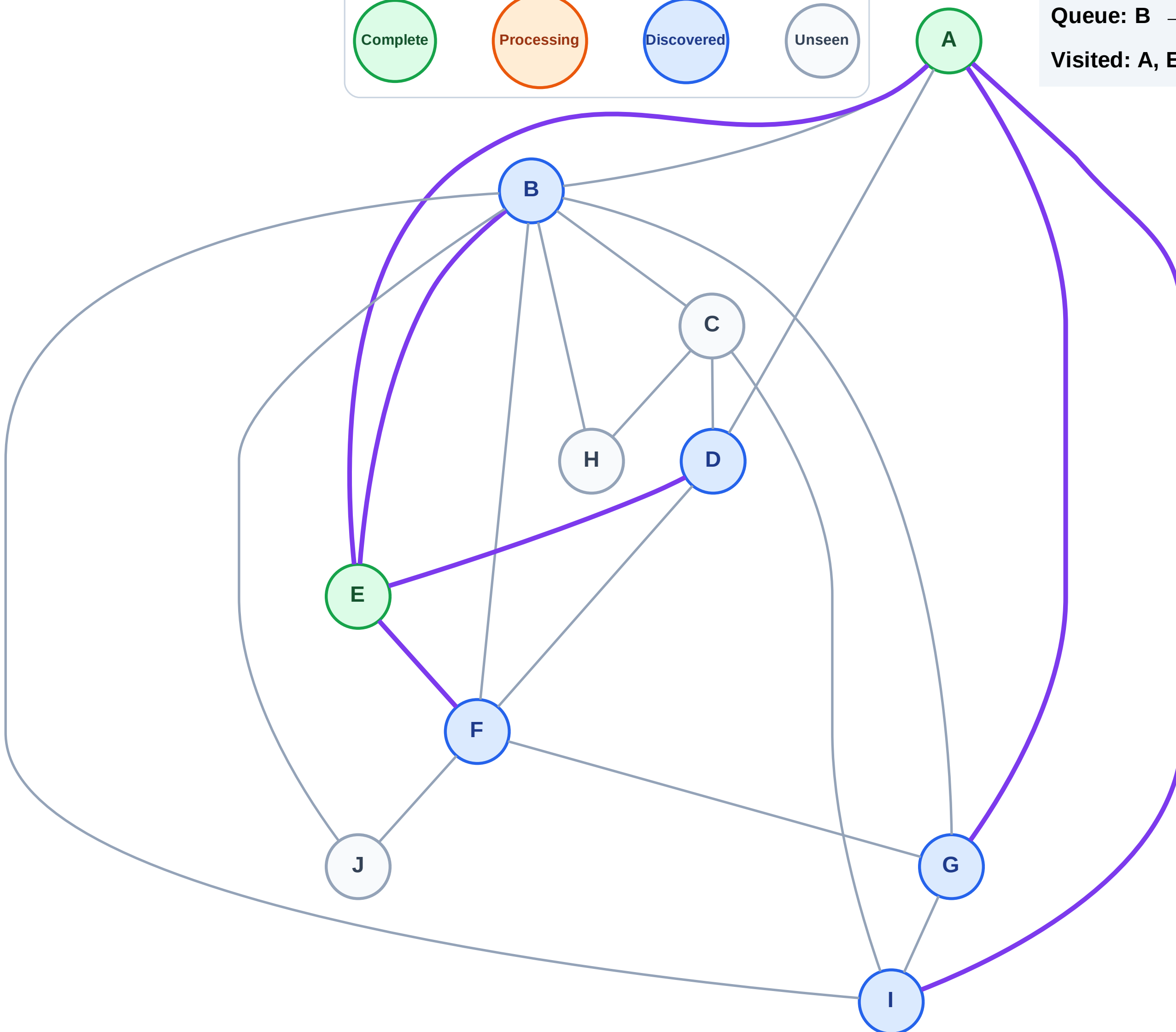
Step 5: Discover G, I from A

Legend



Queue: B → D → F → G → I

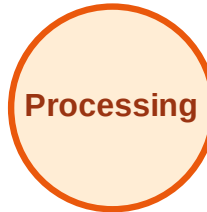
Visited: A, E



BFS Traversal

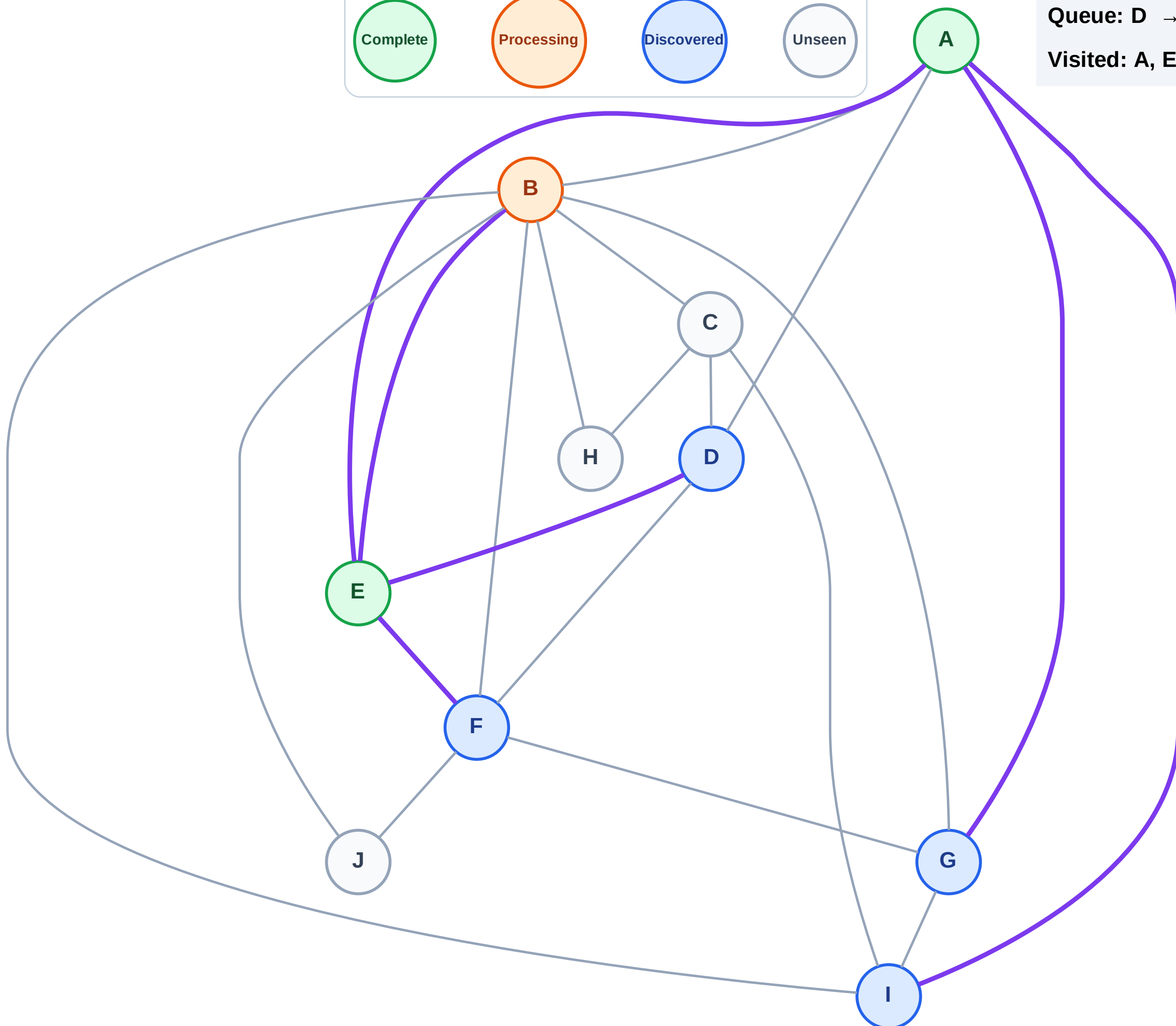
Step 6: Process node B

Legend



Queue: D → F → G → I

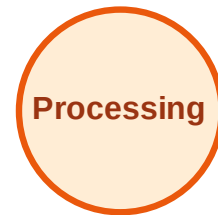
Visited: A, E



BFS Traversal

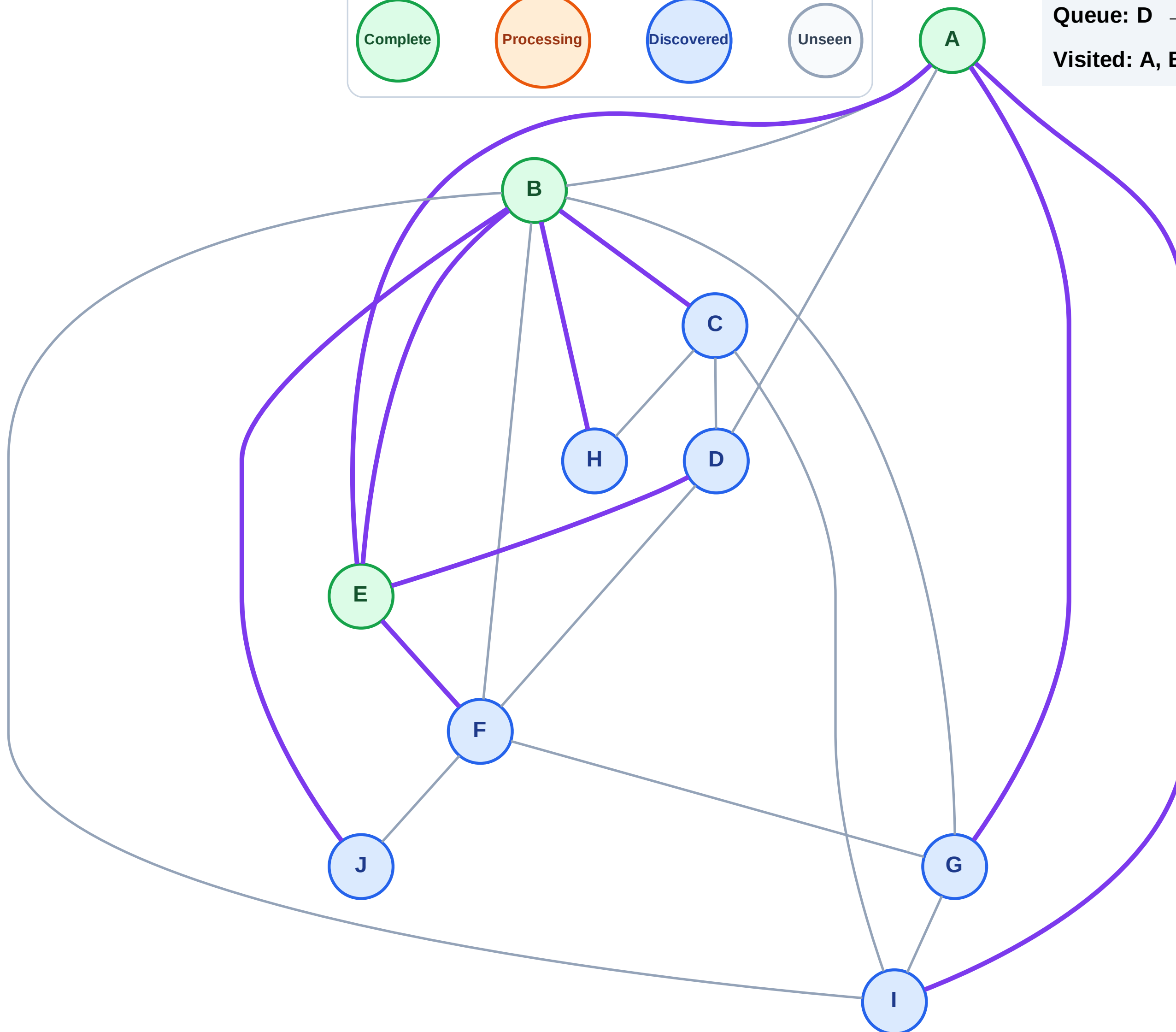
Step 7: Discover C, H, J from B

Legend



Queue: D → F → G → I → C → H → J

Visited: A, B, E



BFS Traversal

Step 8: Process node D

Legend

Complete

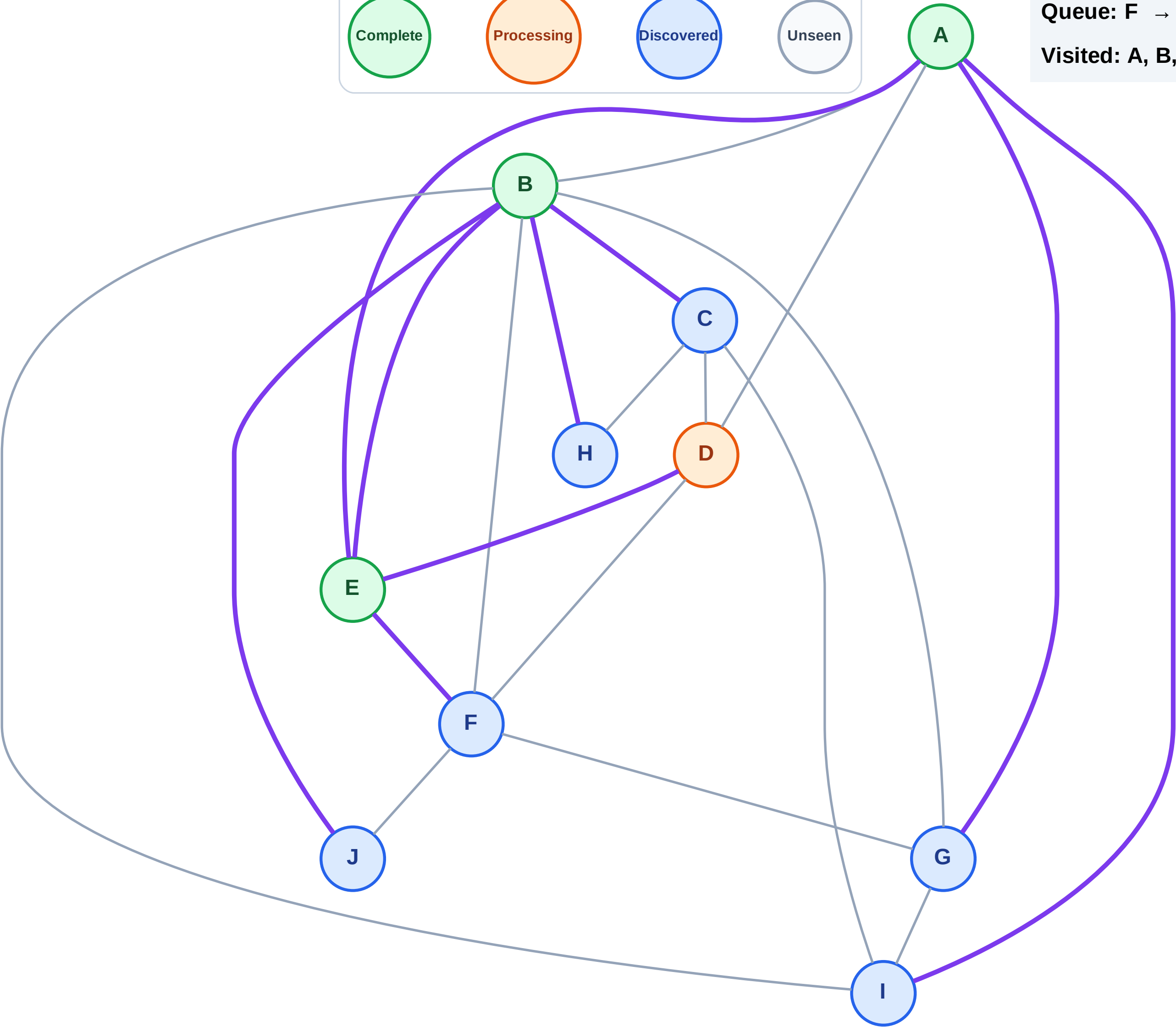
Processing

Discovered

Unseen

Queue: F → G → I → C → H → J

Visited: A, B, E



BFS Traversal

Step 9: No new neighbors from D

Legend

Complete

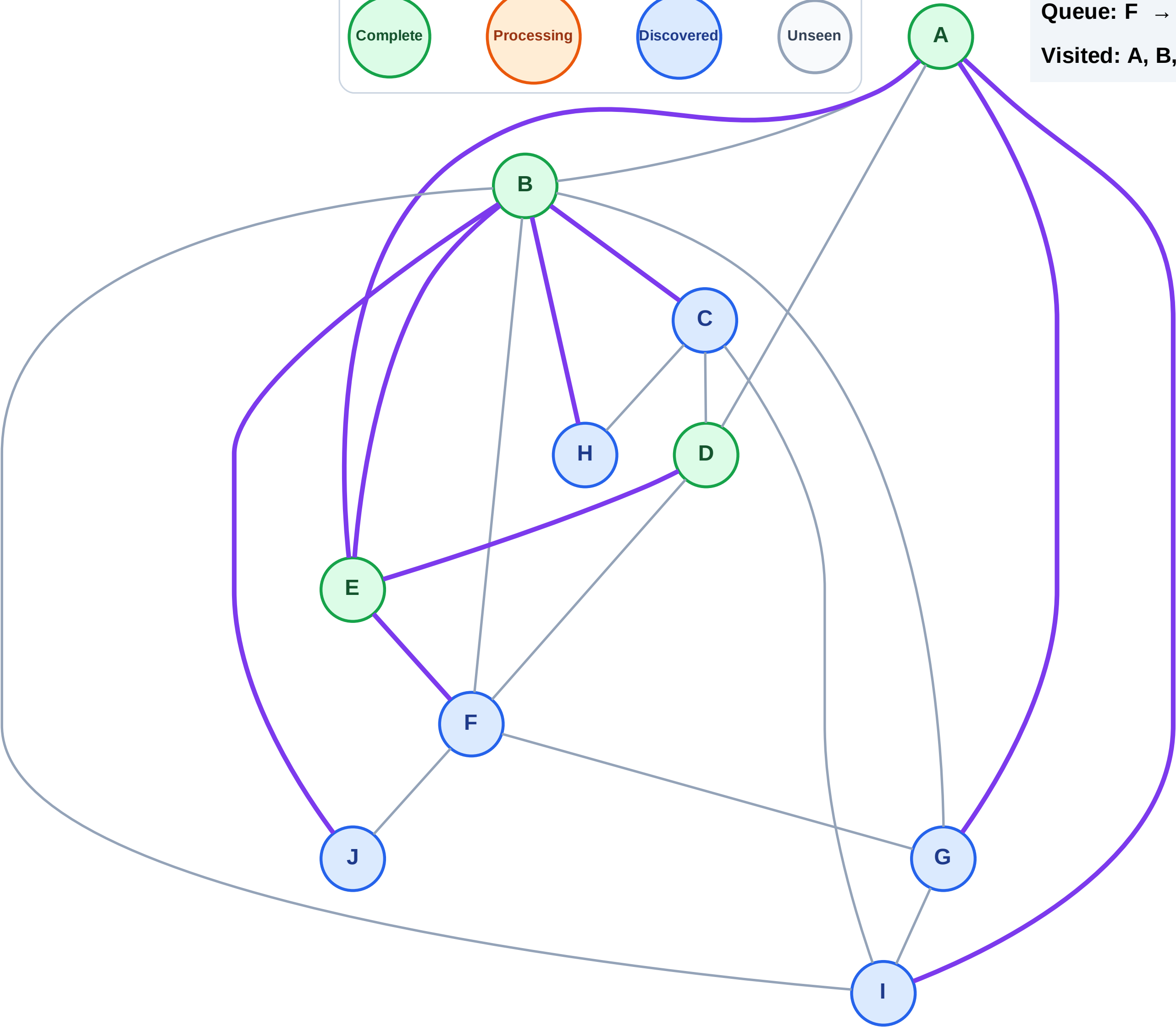
Processing

Discovered

Unseen

Queue: F → G → I → C → H → J

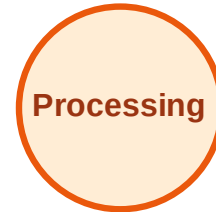
Visited: A, B, D, E



BFS Traversal

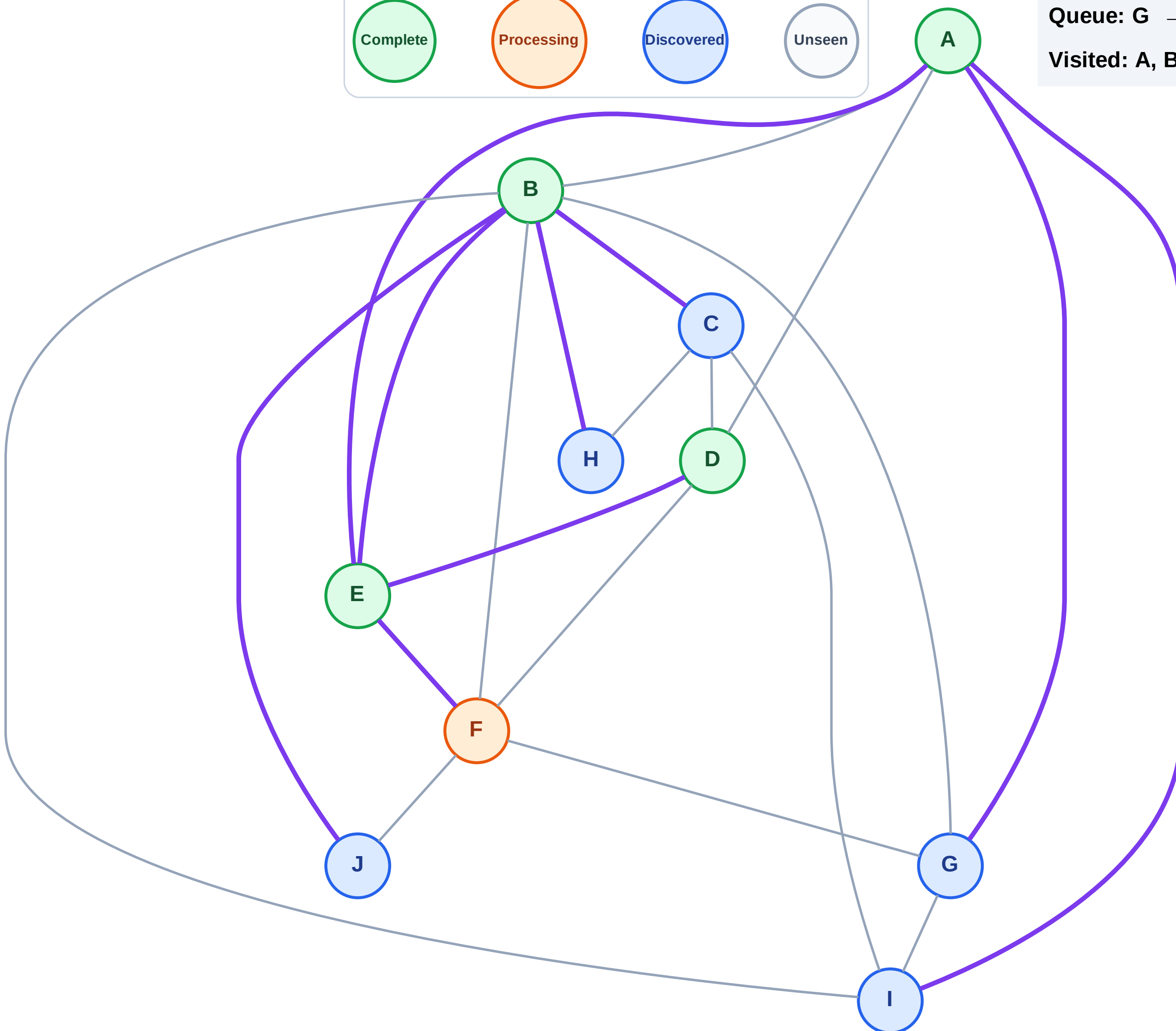
Step 10: Process node F

Legend



Queue: G → I → C → H → J

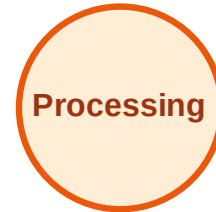
Visited: A, B, D, E



BFS Traversal

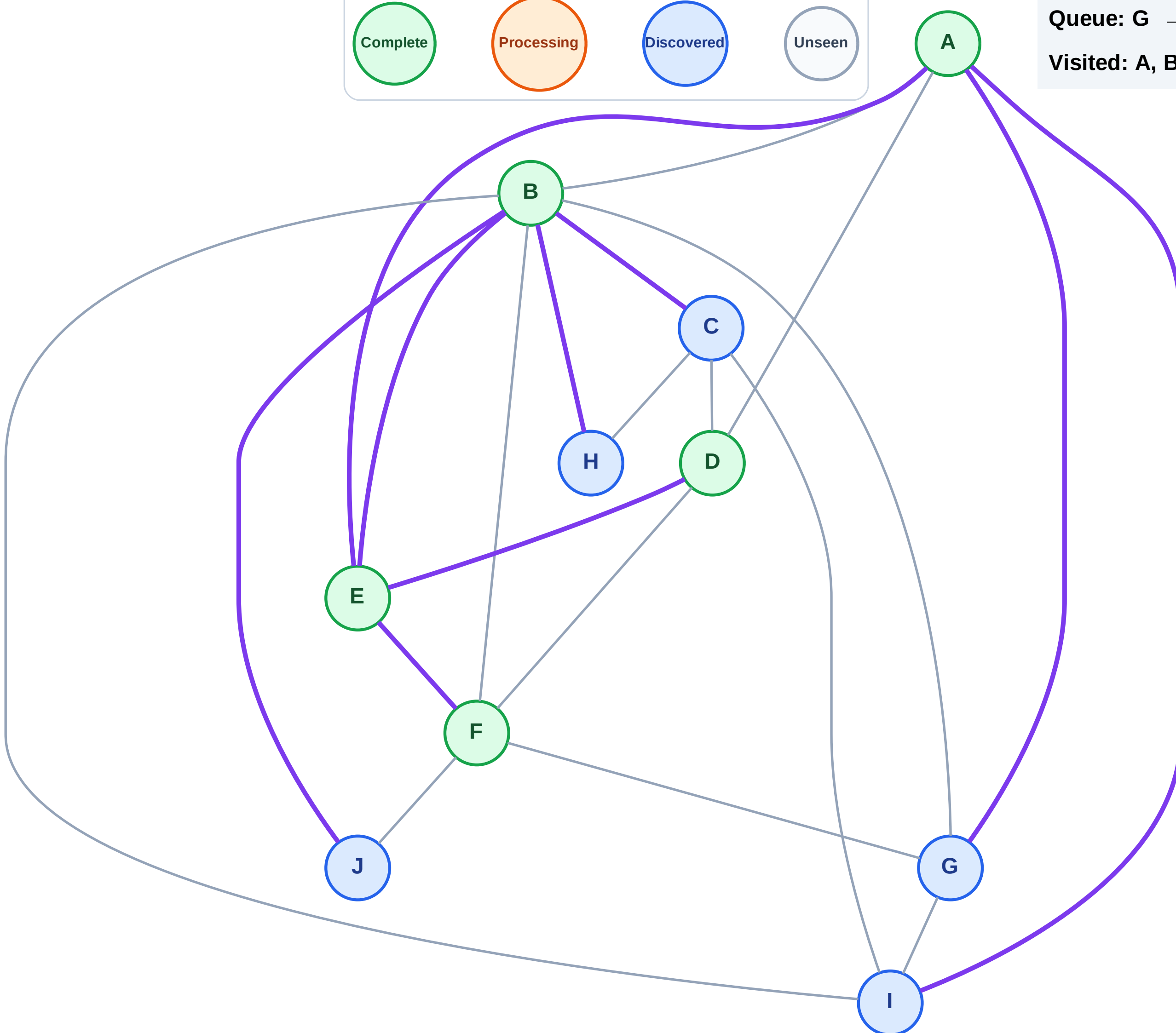
Step 11: No new neighbors from F

Legend



Queue: G → I → C → H → J

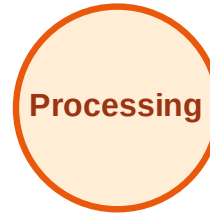
Visited: A, B, D, E, F



BFS Traversal

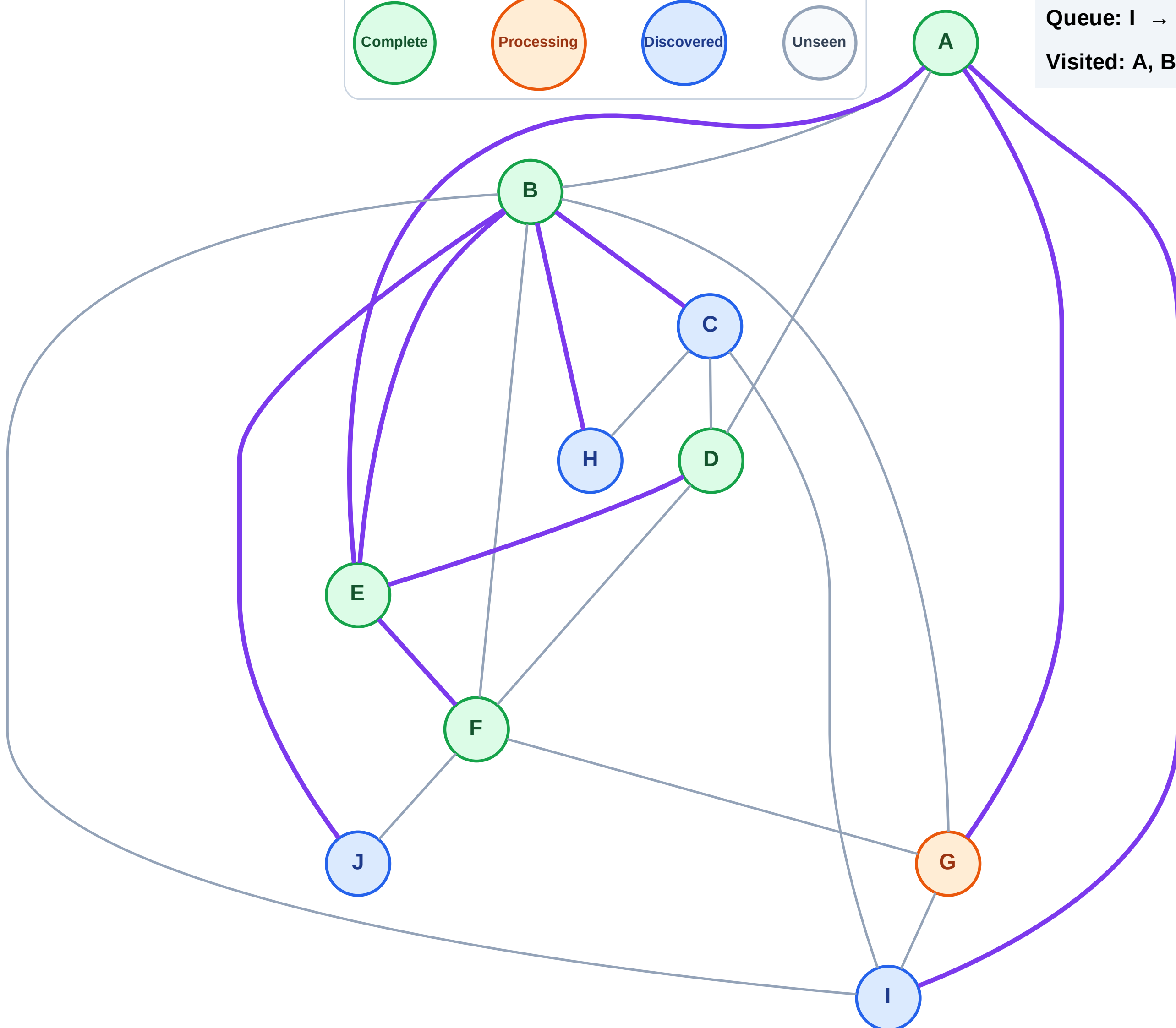
Step 12: Process node G

Legend



Queue: I → C → H → J

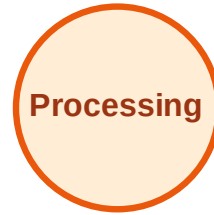
Visited: A, B, D, E, F



BFS Traversal

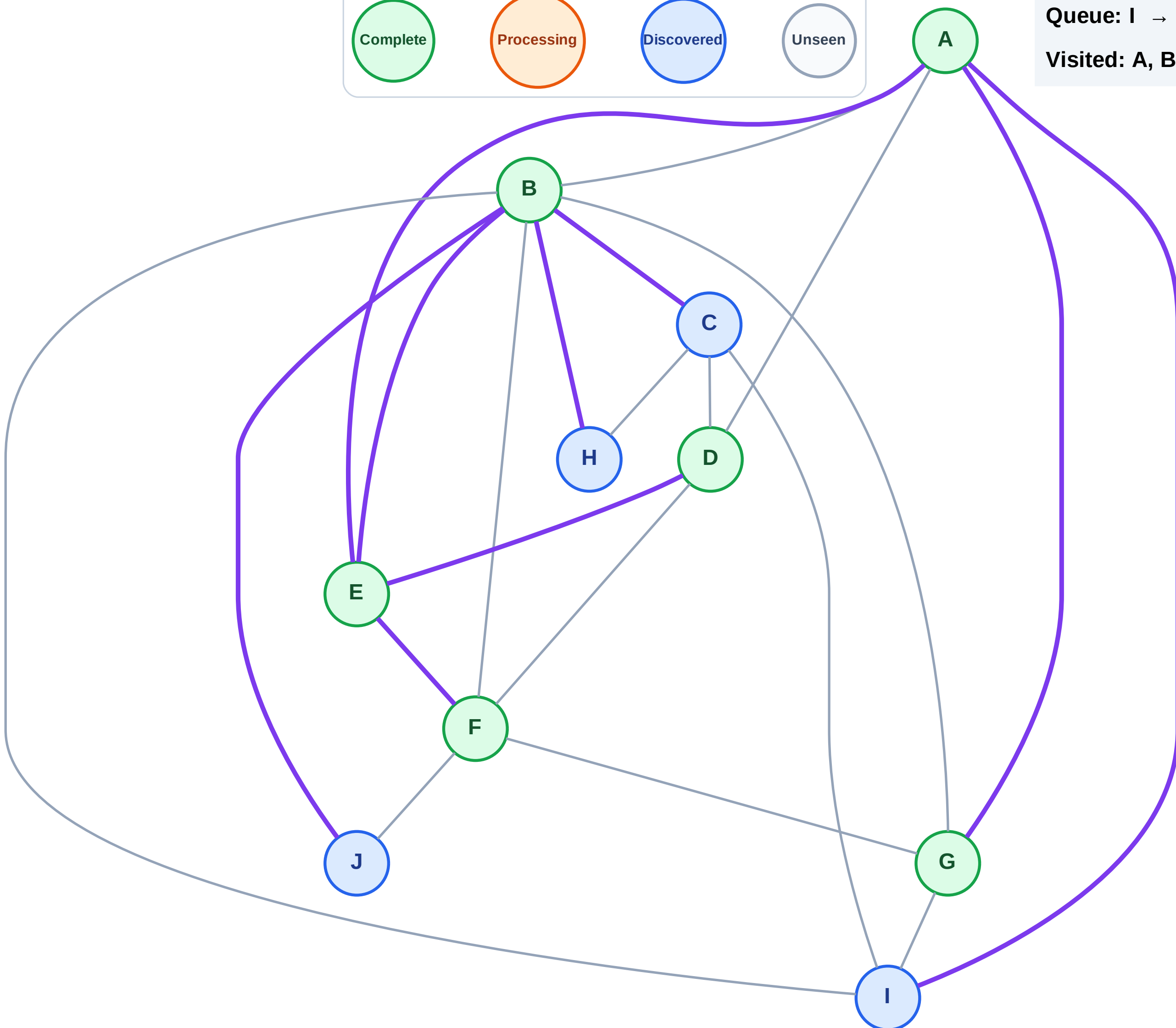
Step 13: No new neighbors from G

Legend



Queue: I → C → H → J

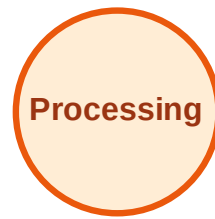
Visited: A, B, D, E, F, G



BFS Traversal

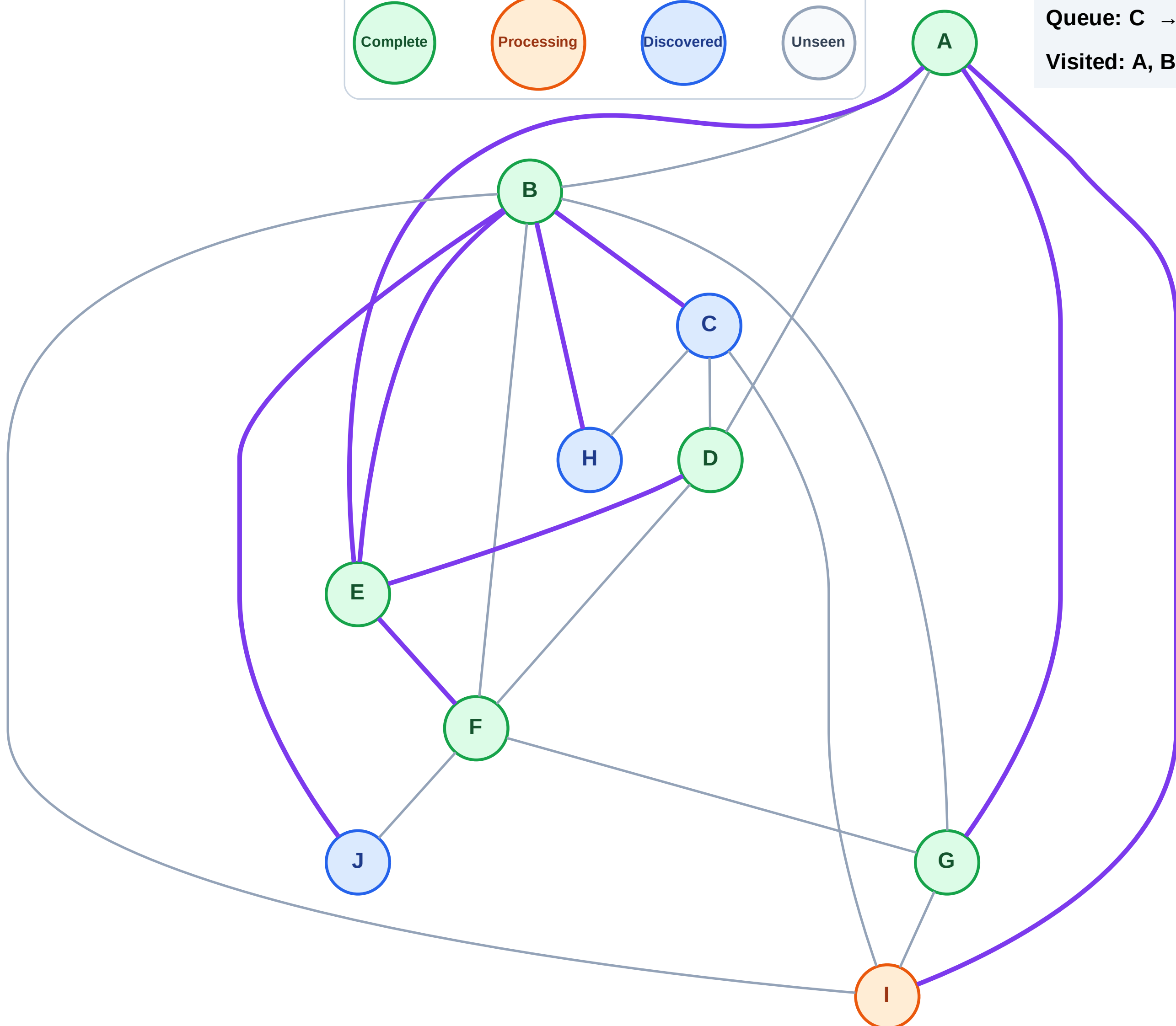
Step 14: Process node I

Legend



Queue: C → H → J

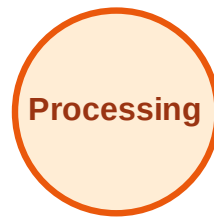
Visited: A, B, D, E, F, G



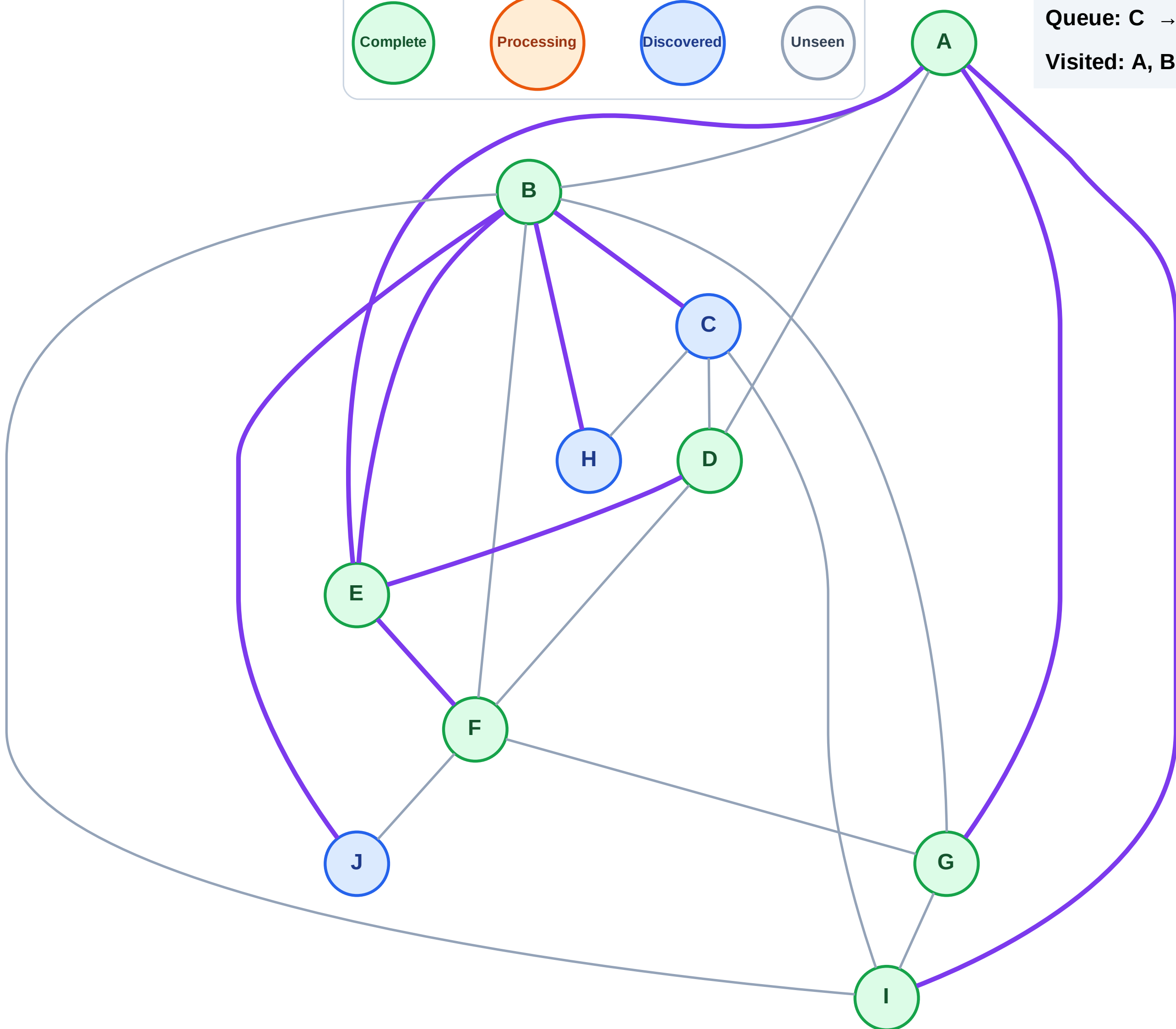
BFS Traversal

Step 15: No new neighbors from I

Legend



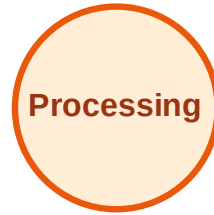
Queue: C → H → J
Visited: A, B, D, E, F, G, I



BFS Traversal

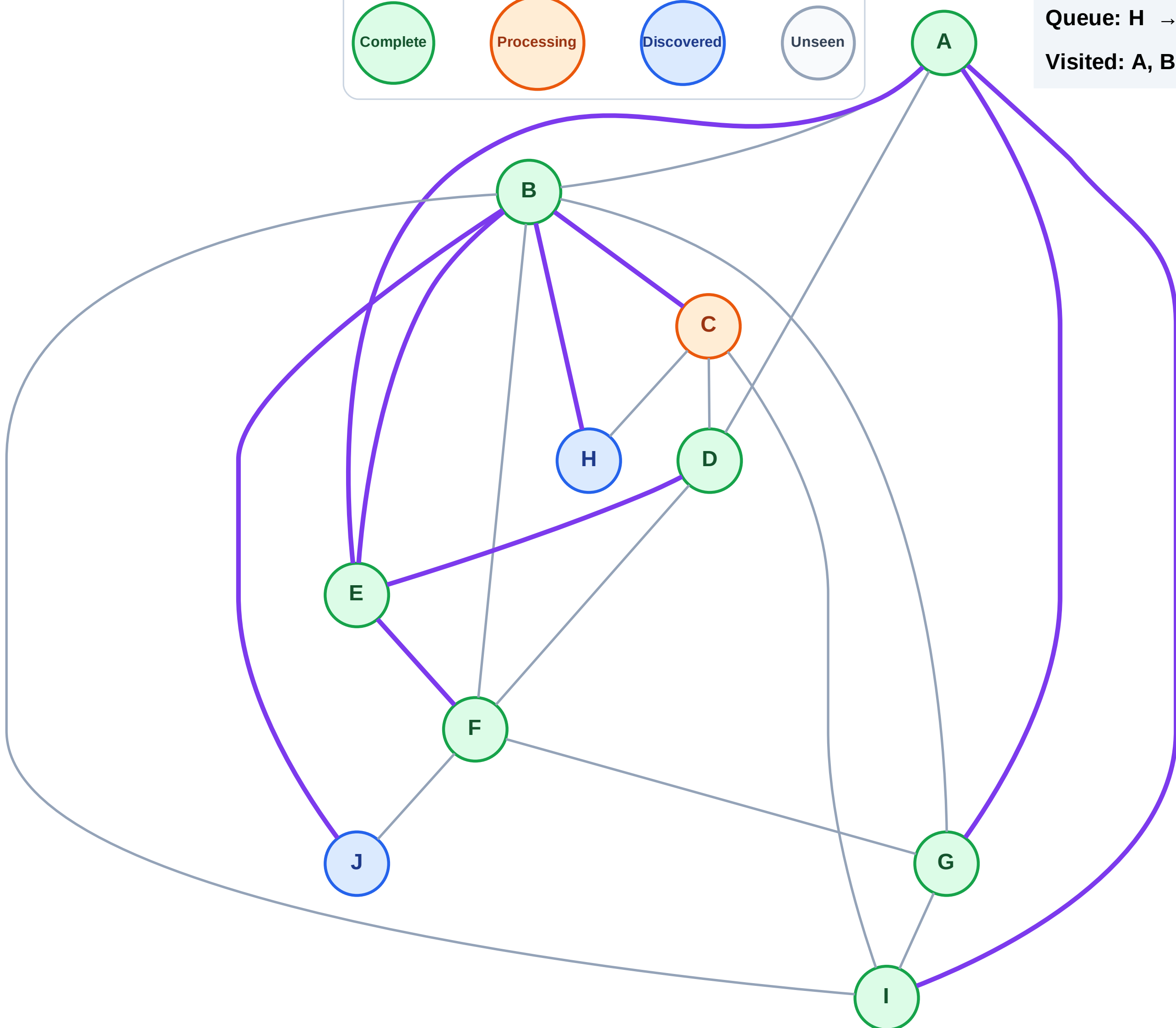
Step 16: Process node C

Legend



Queue: H → J

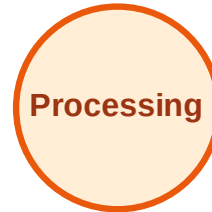
Visited: A, B, D, E, F, G, I



BFS Traversal

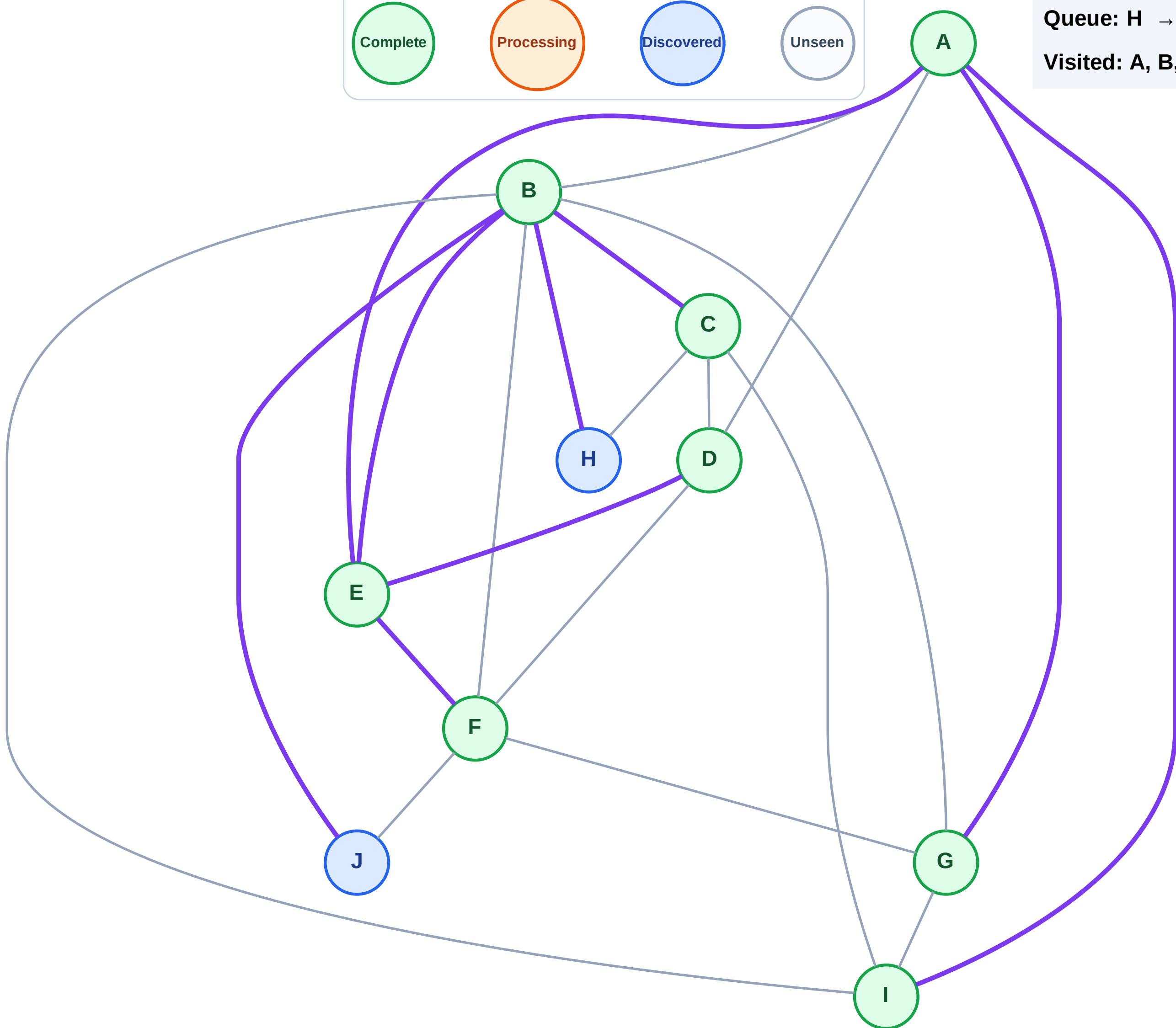
Step 17: No new neighbors from C

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

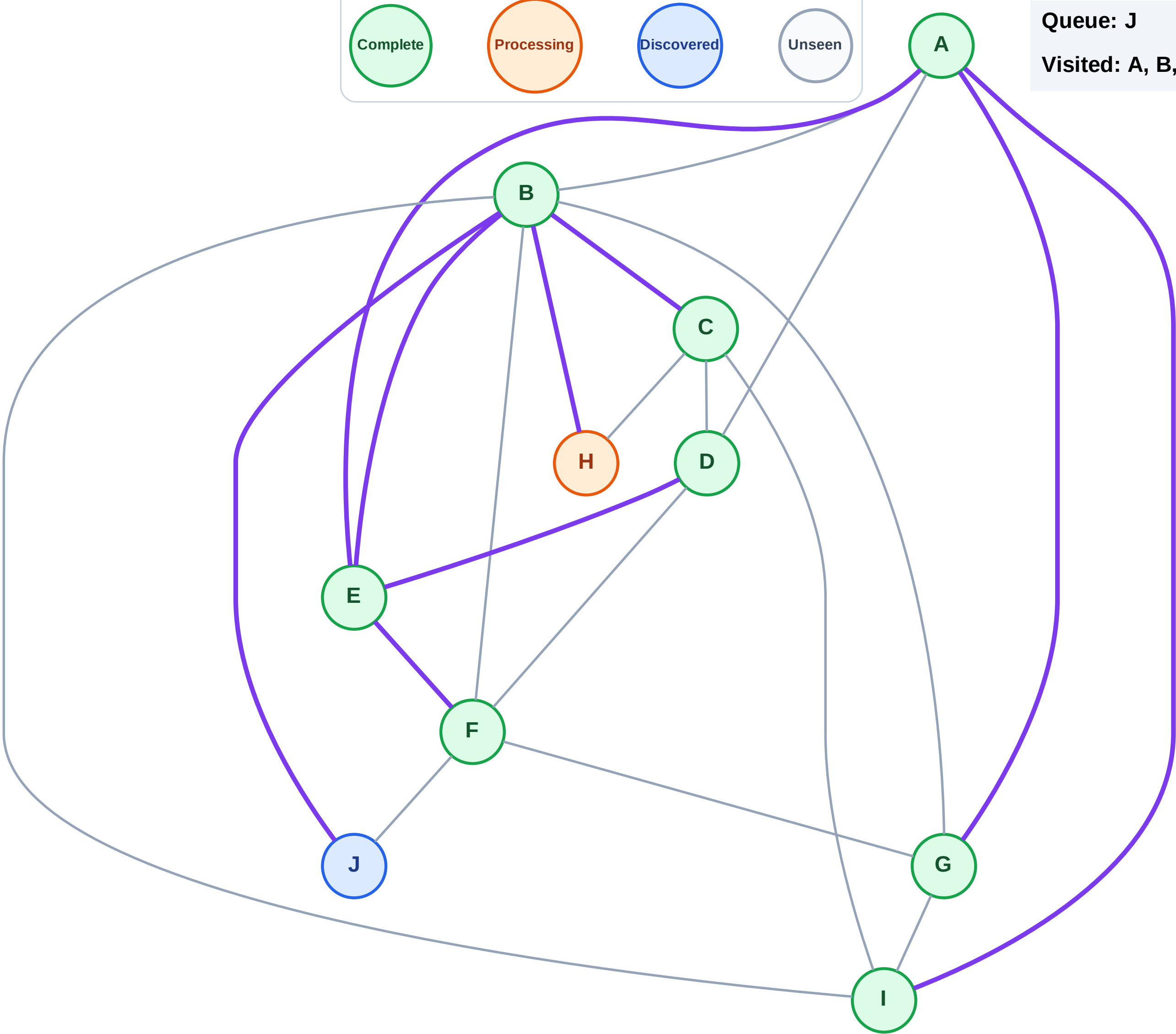
Step 18: Process node H

Legend



Queue: J

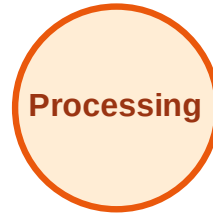
Visited: A, B, C, D, E, F, G, I



BFS Traversal

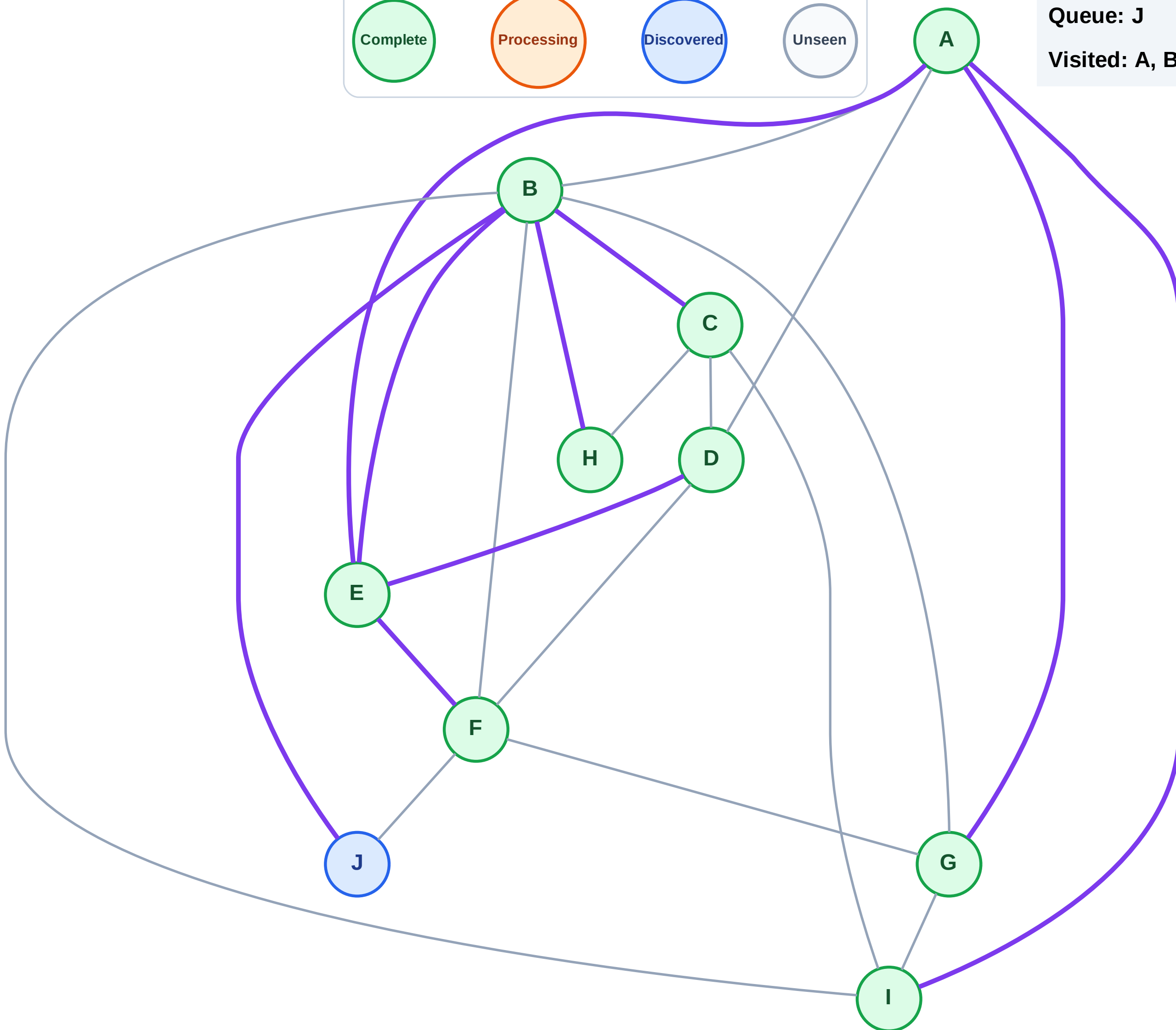
Step 19: No new neighbors from H

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend



Queue: (empty)

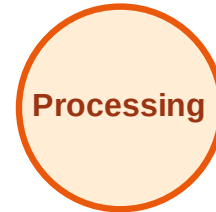
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

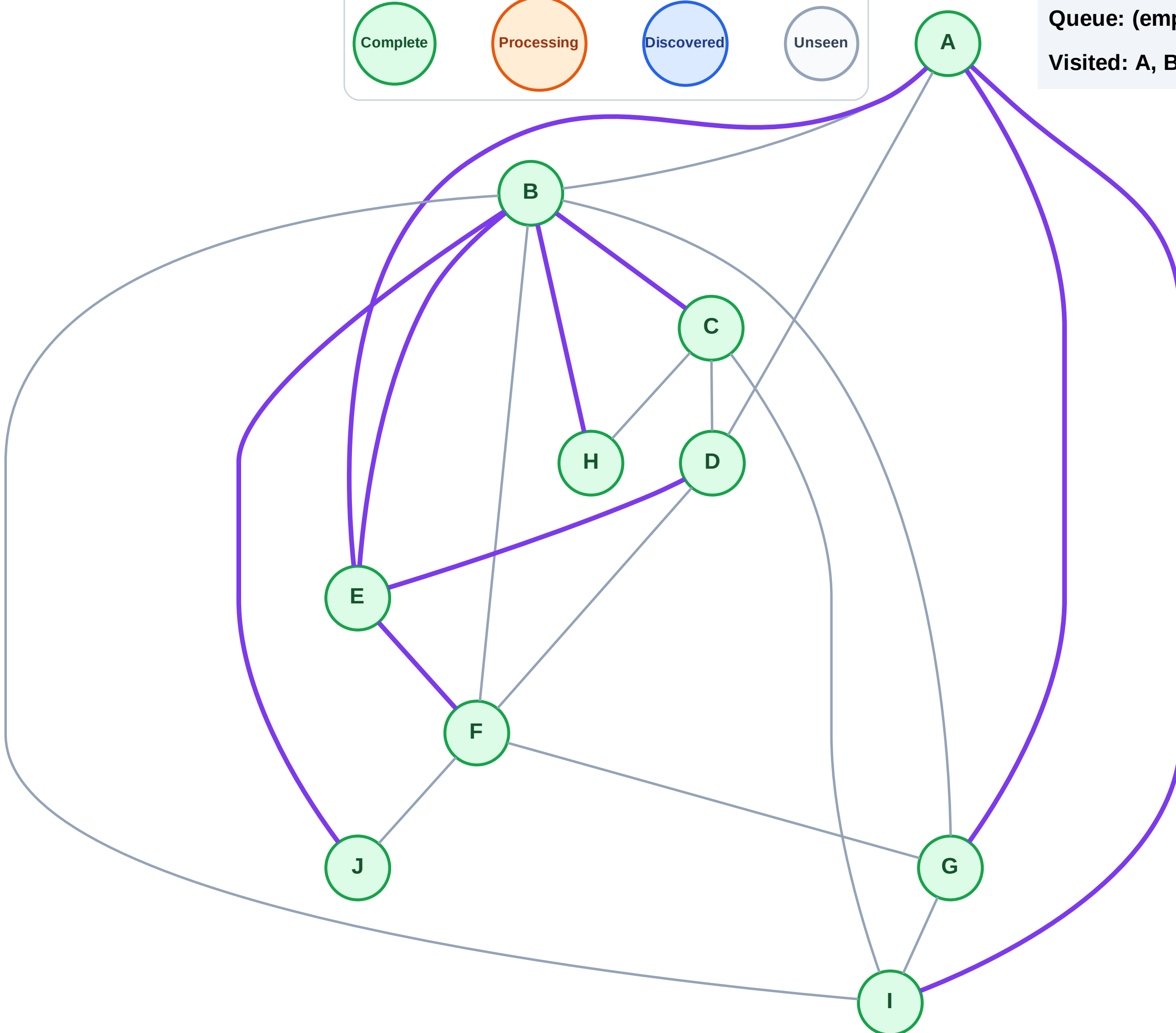
Step 21: No new neighbors from J

Legend



Queue: (empty)

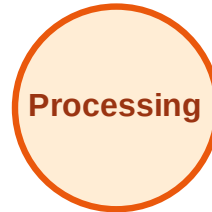
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

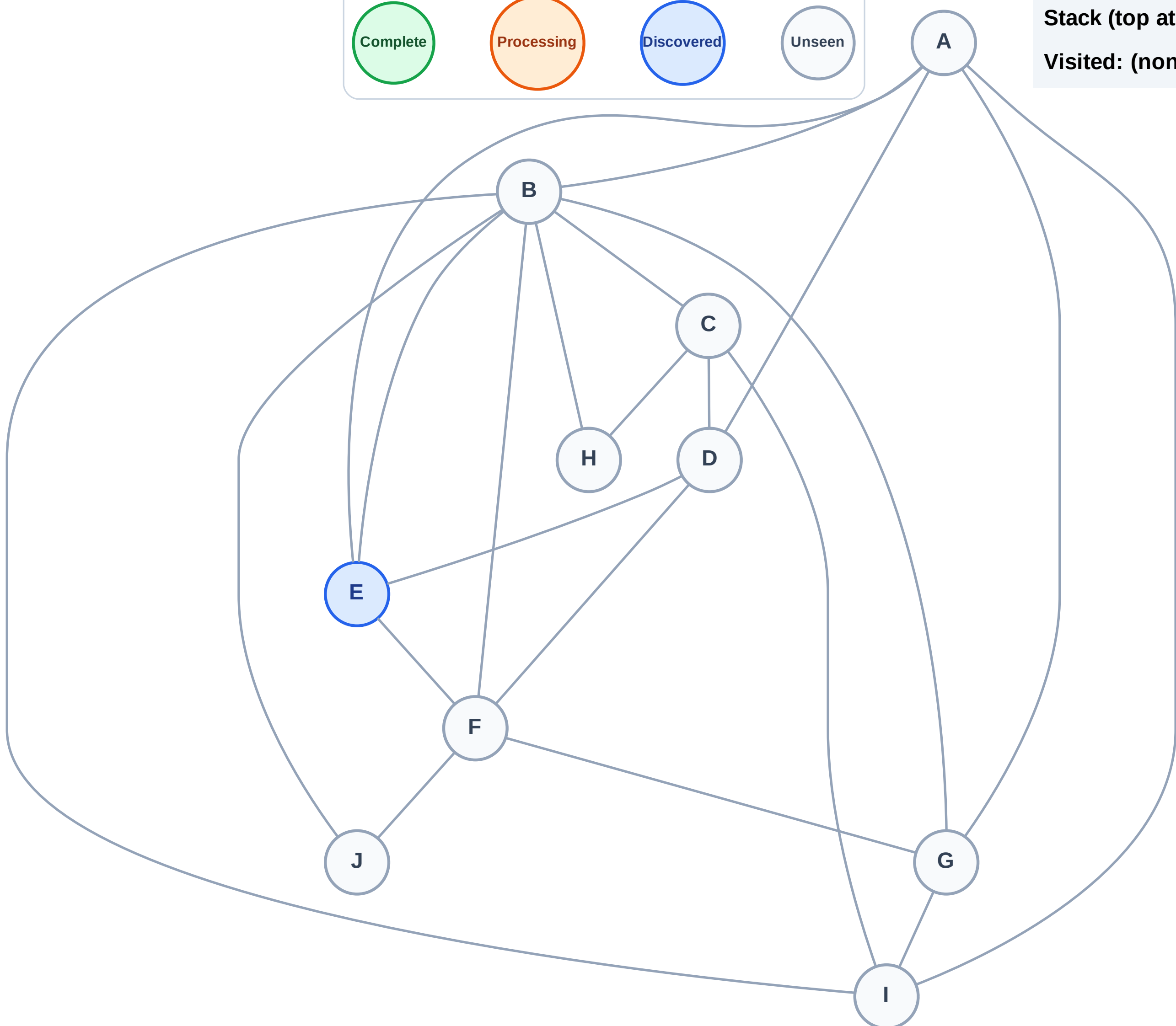
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



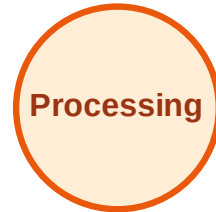
DFS Traversal

Step 2: Process node E

Legend



Complete



Processing



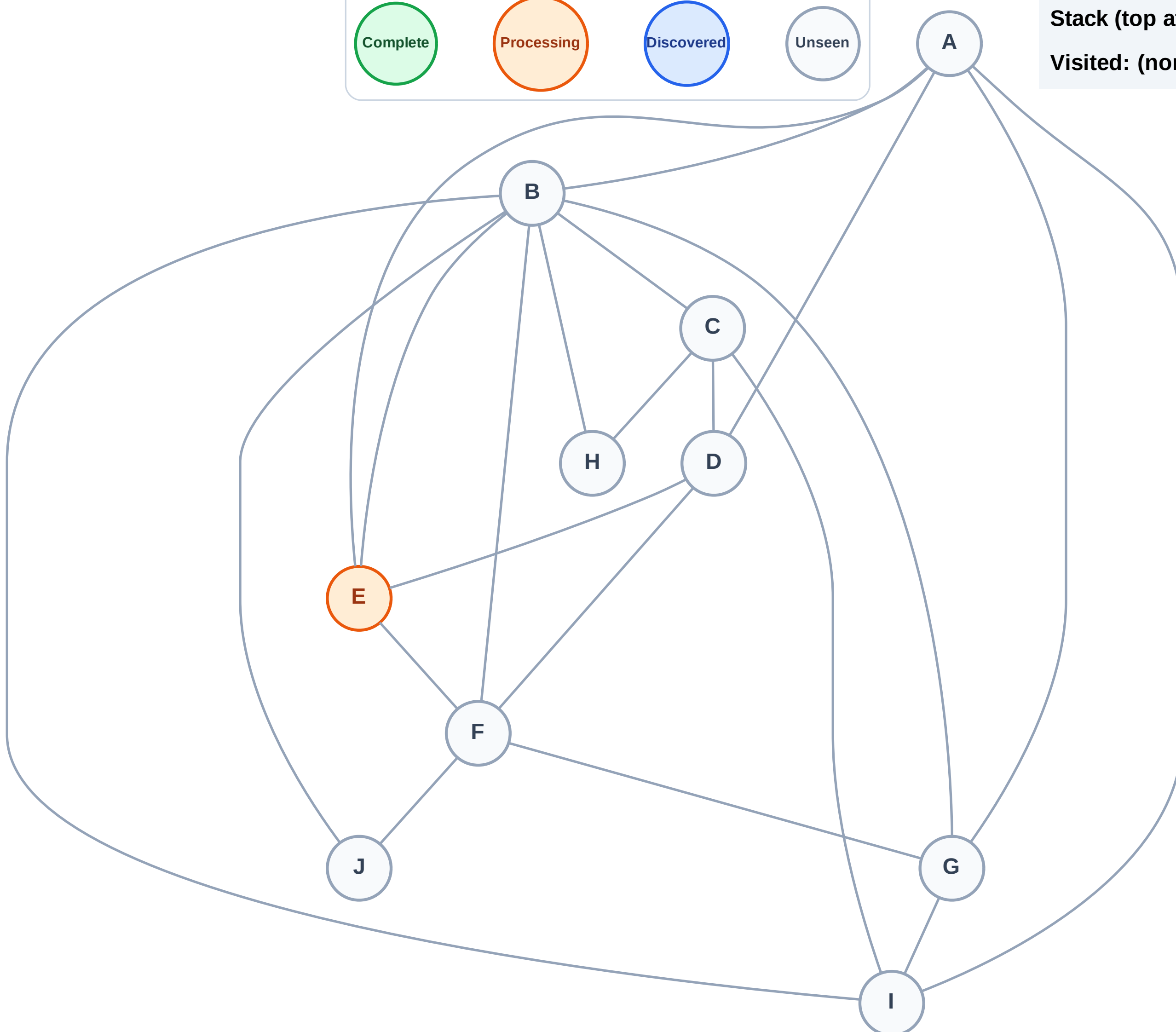
Discovered



Unseen

Stack (top at right): (empty)

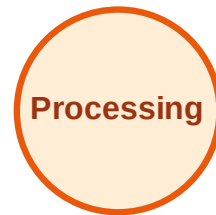
Visited: (none)



DFS Traversal

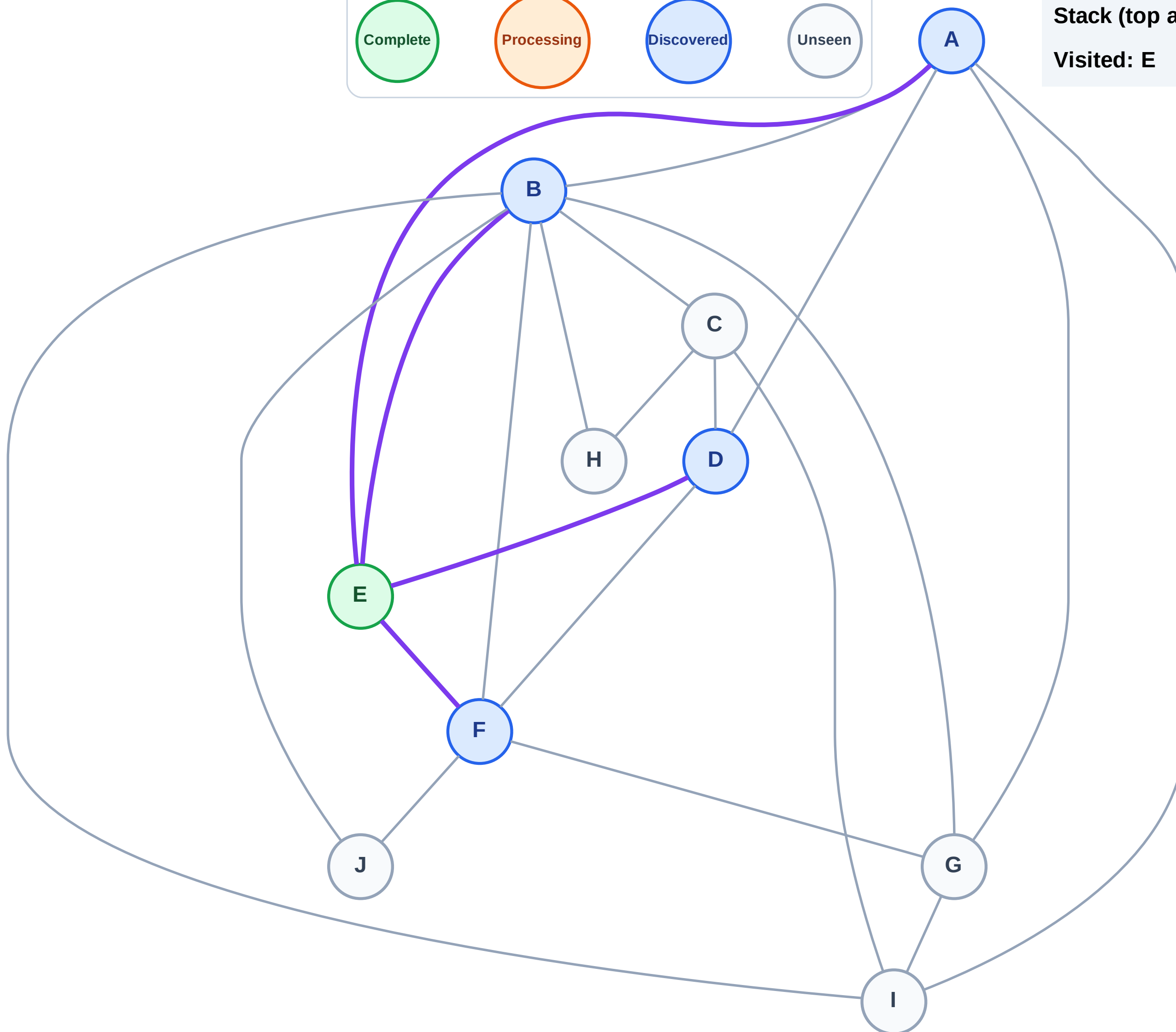
Step 3: Discover F, D, B, A from E

Legend



Stack (top at right): F | D | B | A

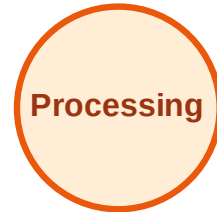
Visited: E



DFS Traversal

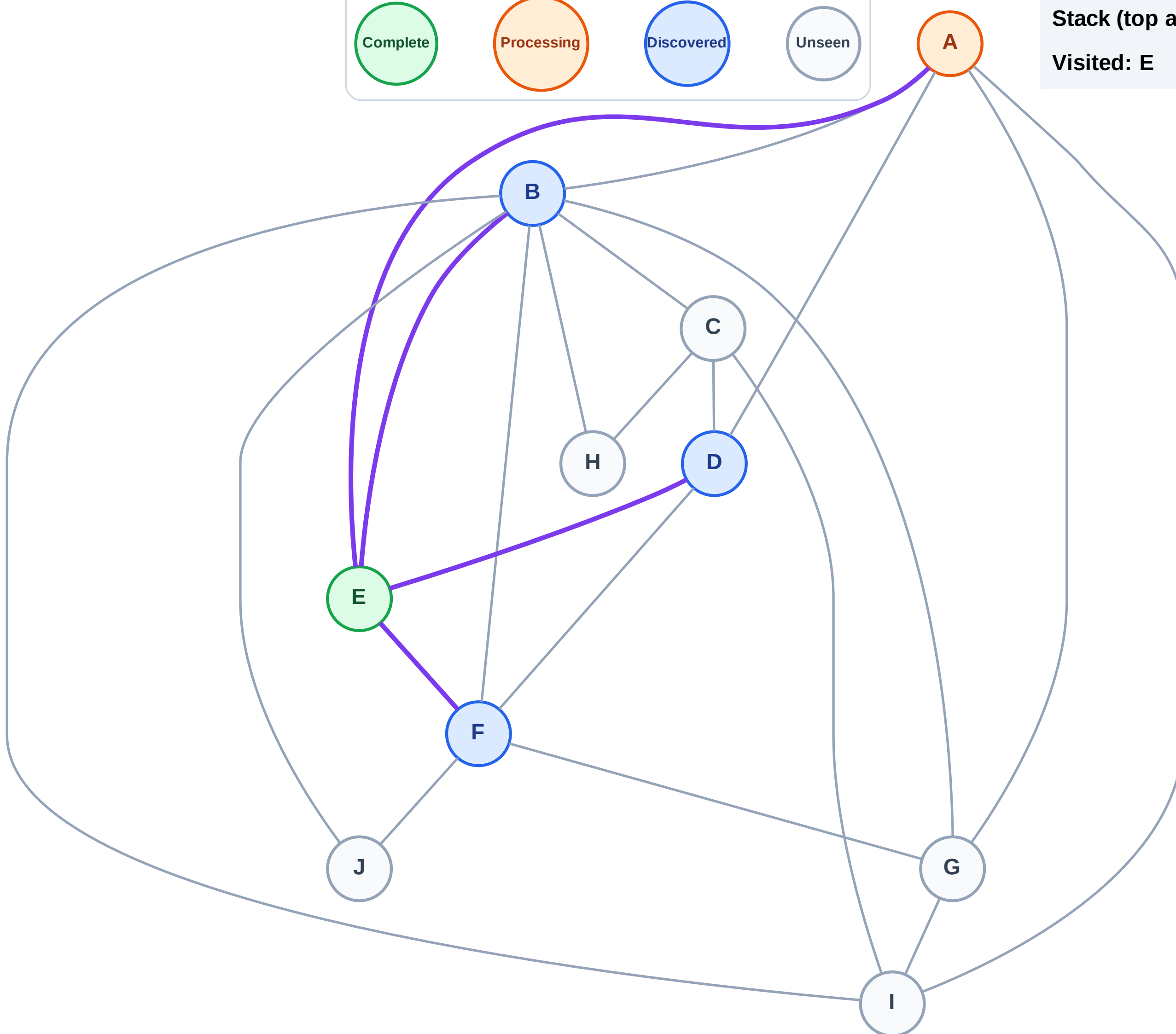
Step 4: Process node A

Legend



Stack (top at right): F | D | B

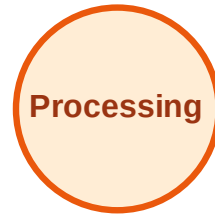
Visited: E



DFS Traversal

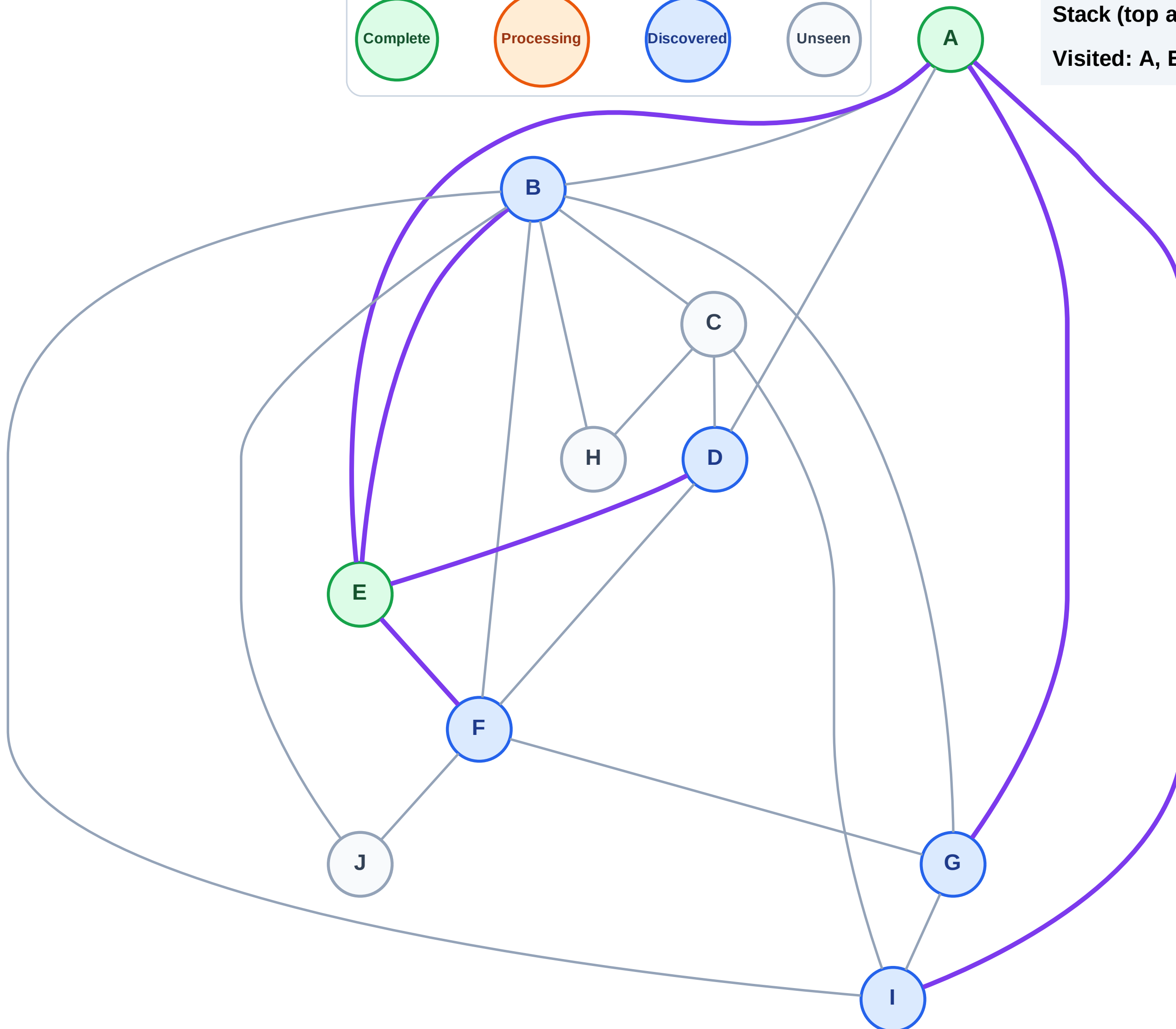
Step 5: Discover I, G from A

Legend



Stack (top at right): F | D | B | I | G

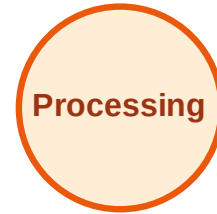
Visited: A, E



DFS Traversal

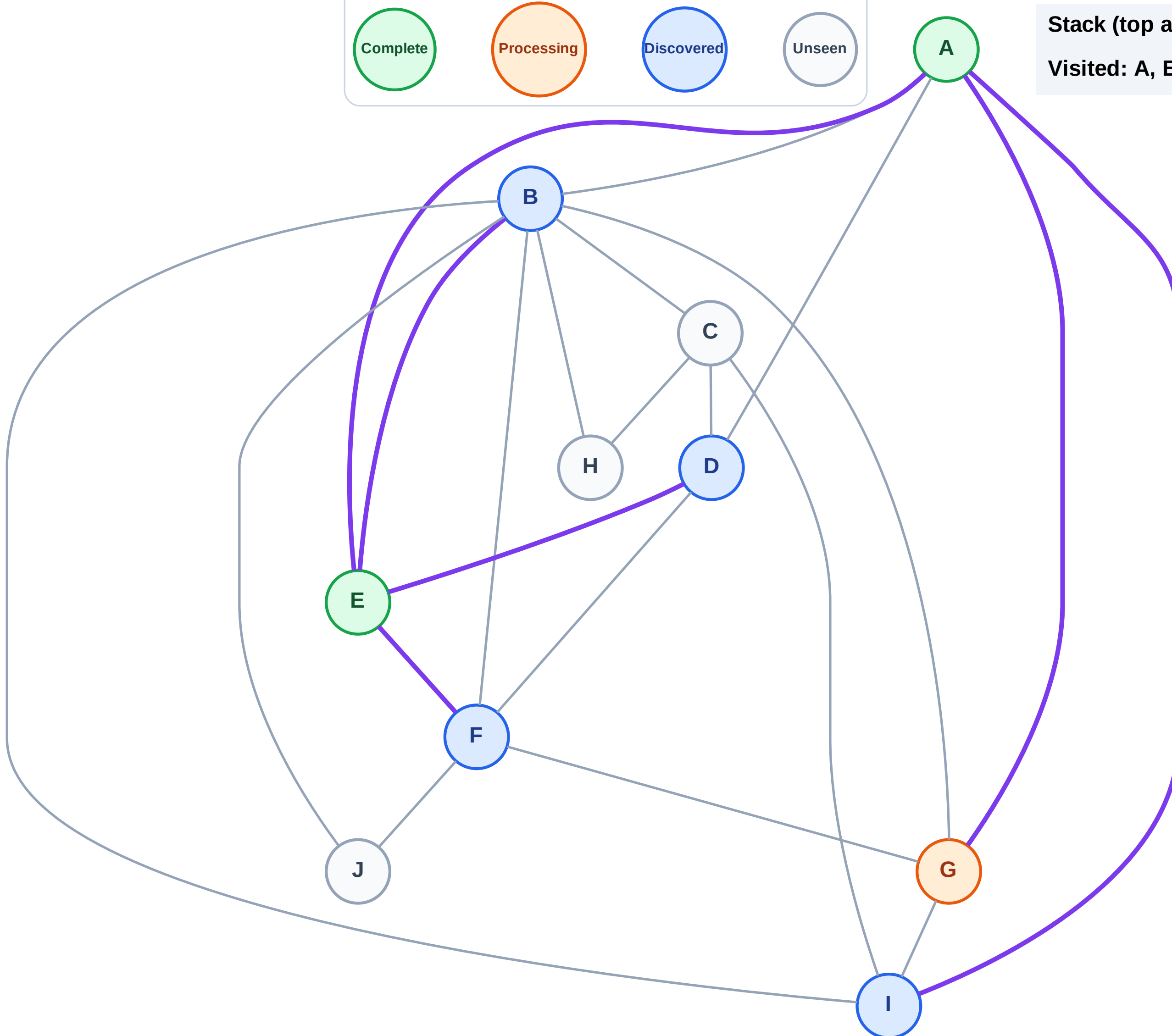
Step 6: Process node G

Legend



Stack (top at right): F | D | B | I

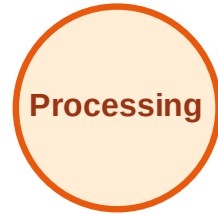
Visited: A, E



DFS Traversal

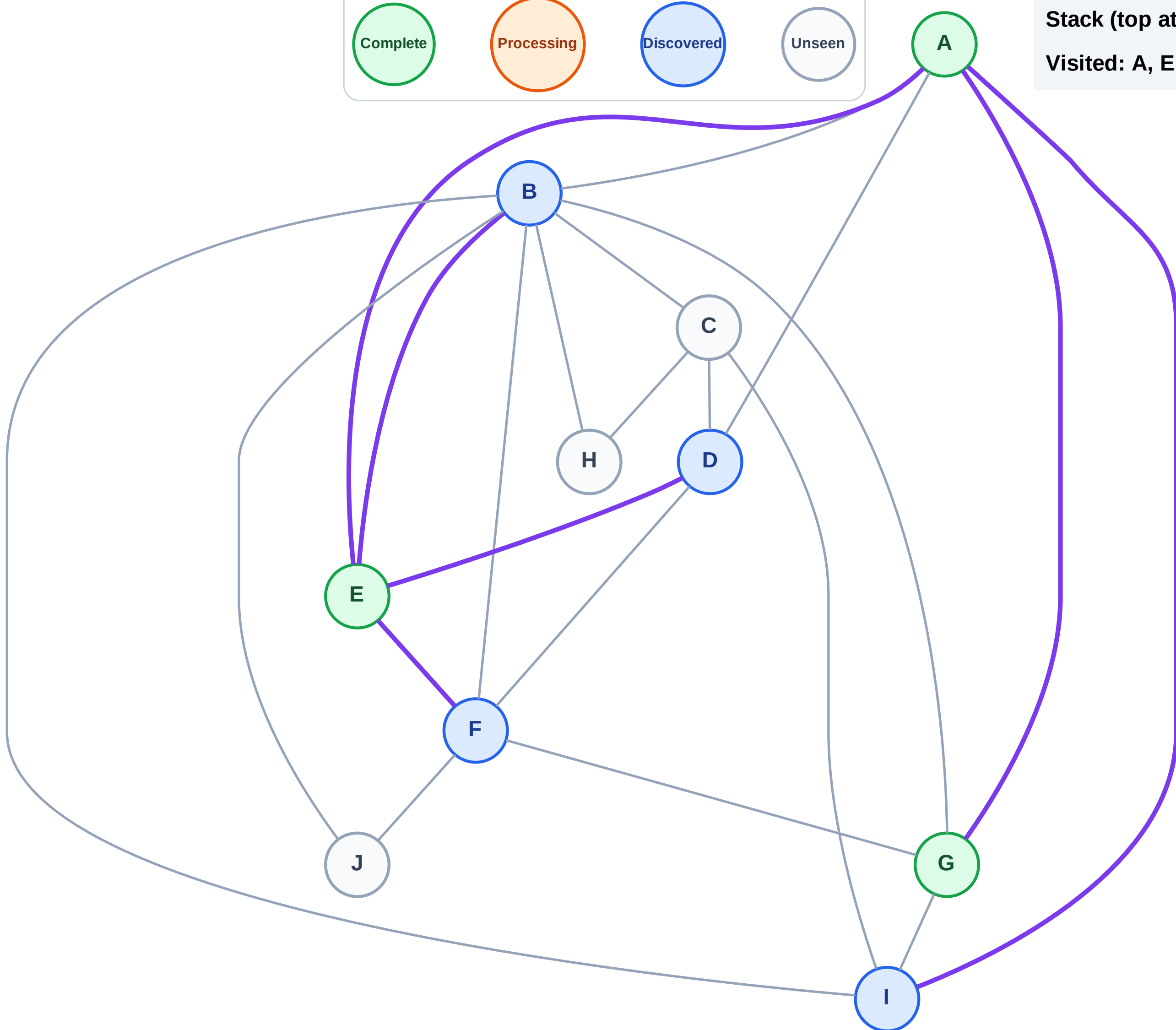
Step 7: No new neighbors from G

Legend



Stack (top at right): F | D | B | I

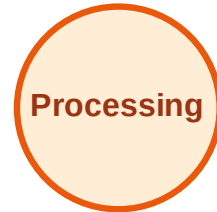
Visited: A, E, G



DFS Traversal

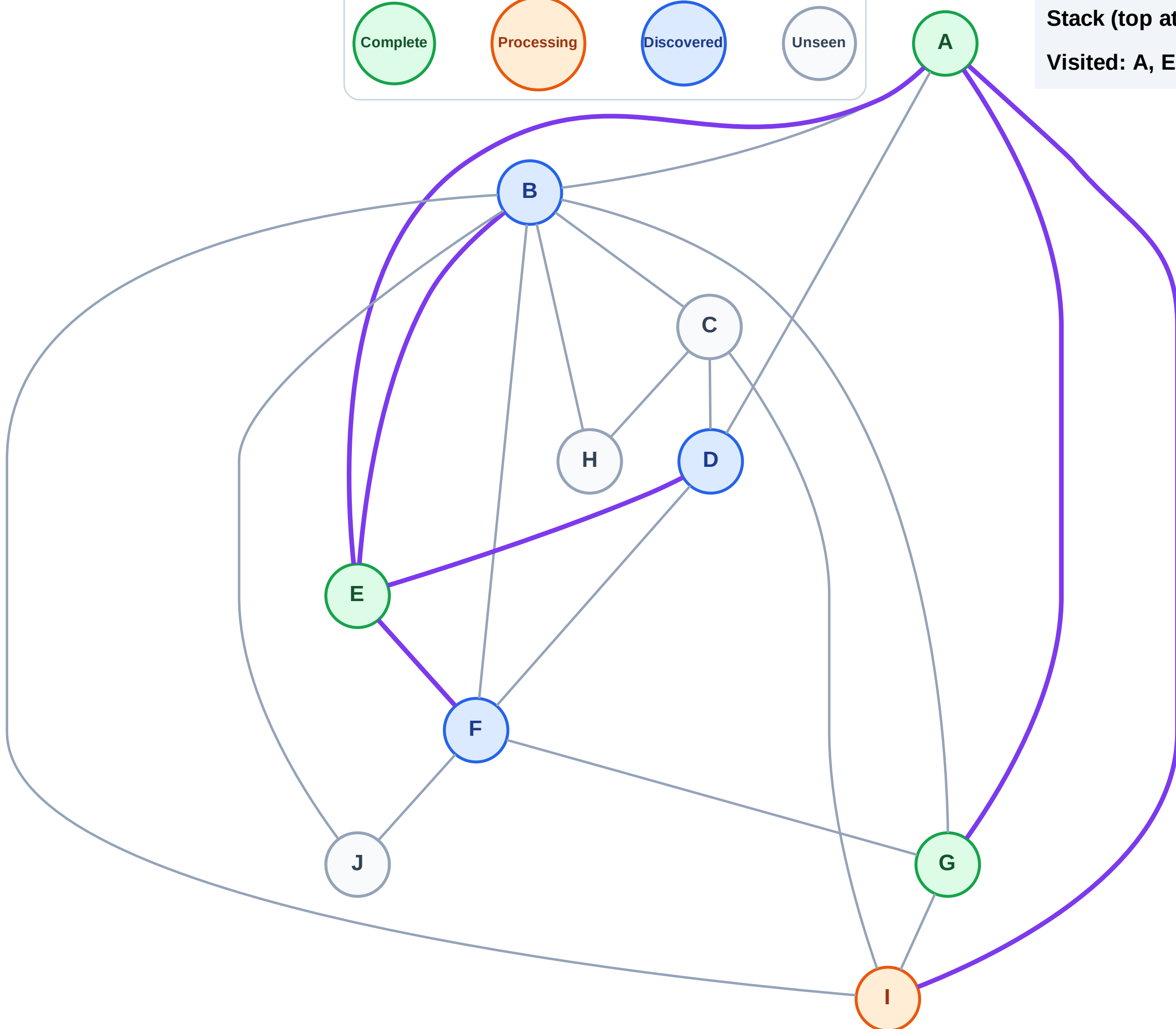
Step 8: Process node I

Legend



Stack (top at right): F | D | B

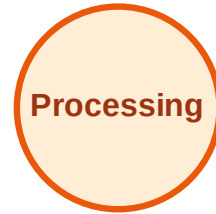
Visited: A, E, G



DFS Traversal

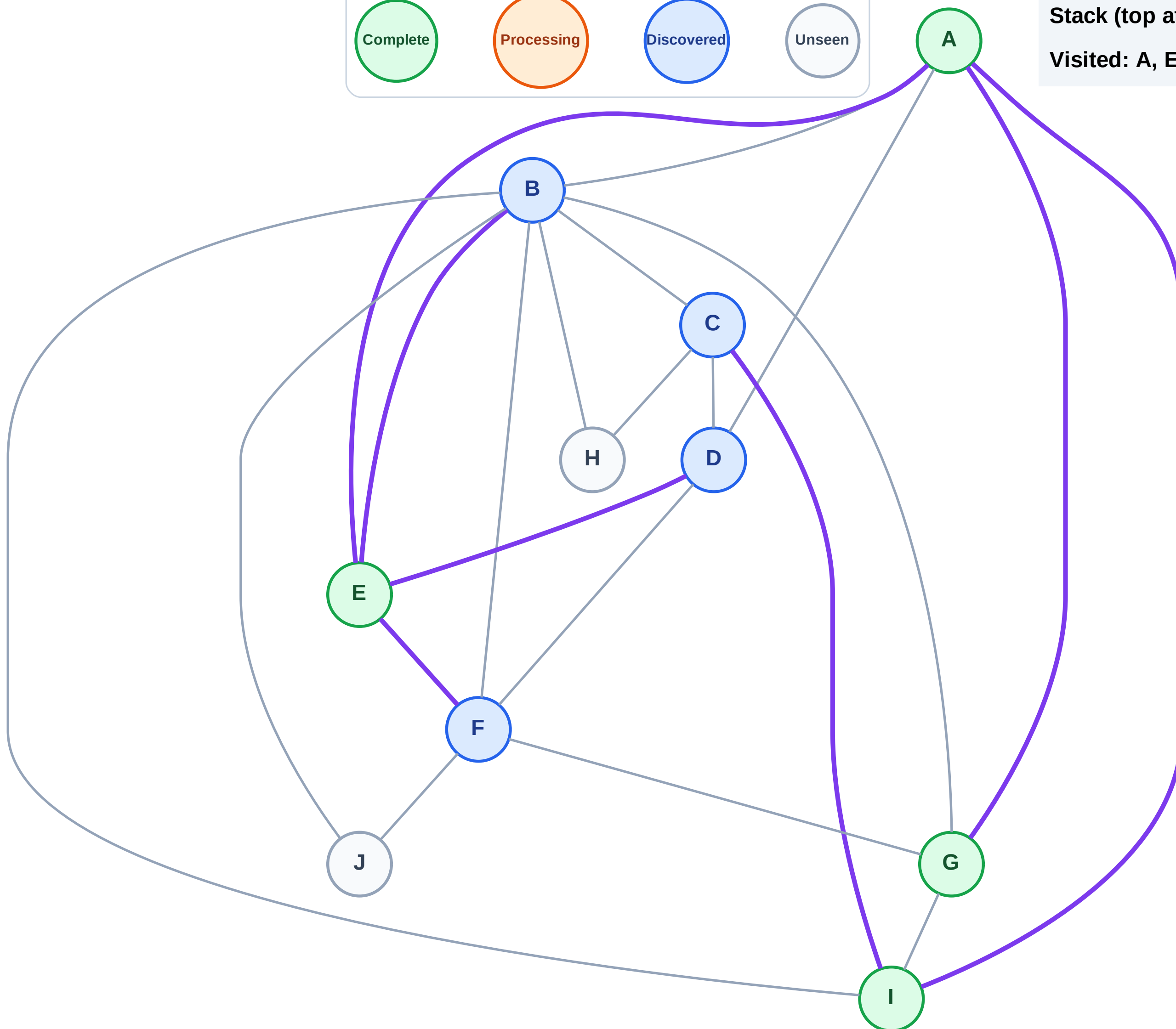
Step 9: Discover C from I

Legend



Stack (top at right): F | D | B | C

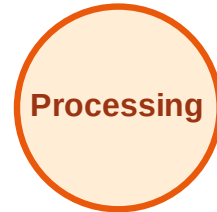
Visited: A, E, G, I



DFS Traversal

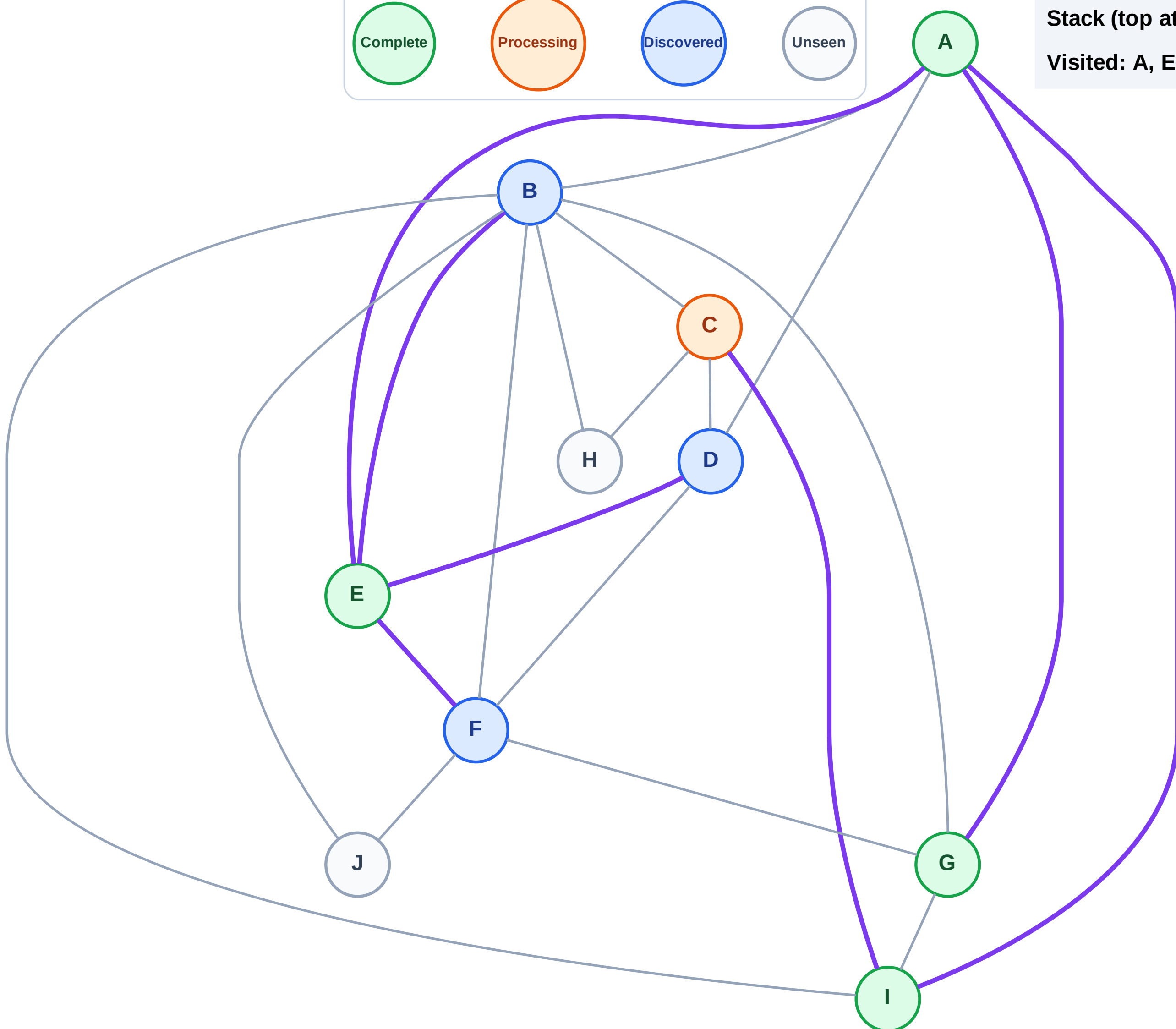
Step 10: Process node C

Legend



Stack (top at right): F | D | B

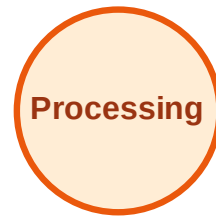
Visited: A, E, G, I



DFS Traversal

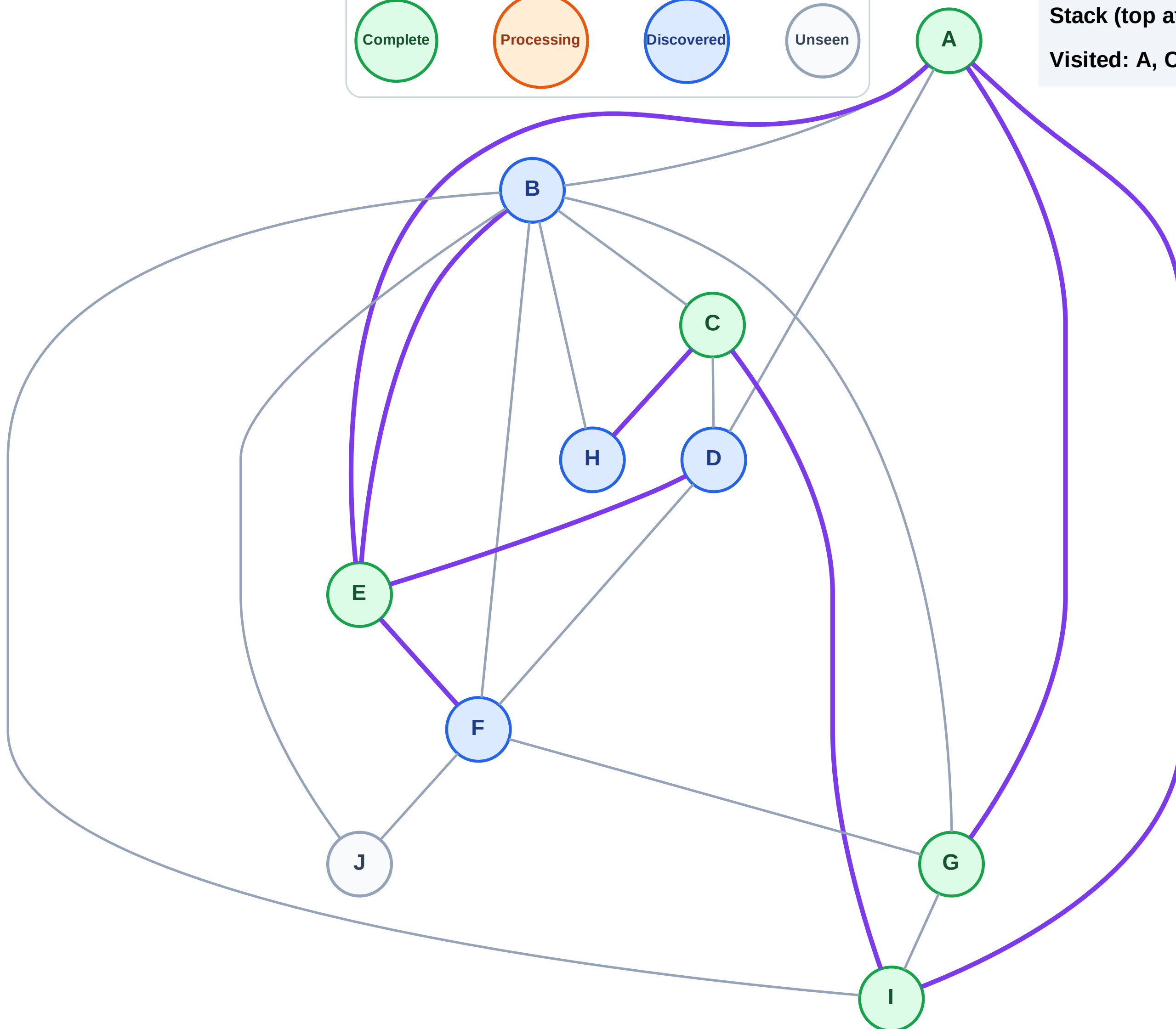
Step 11: Discover H from C

Legend



Stack (top at right): F | D | B | H

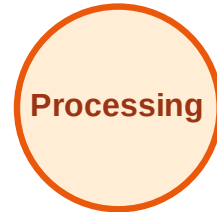
Visited: A, C, E, G, I



DFS Traversal

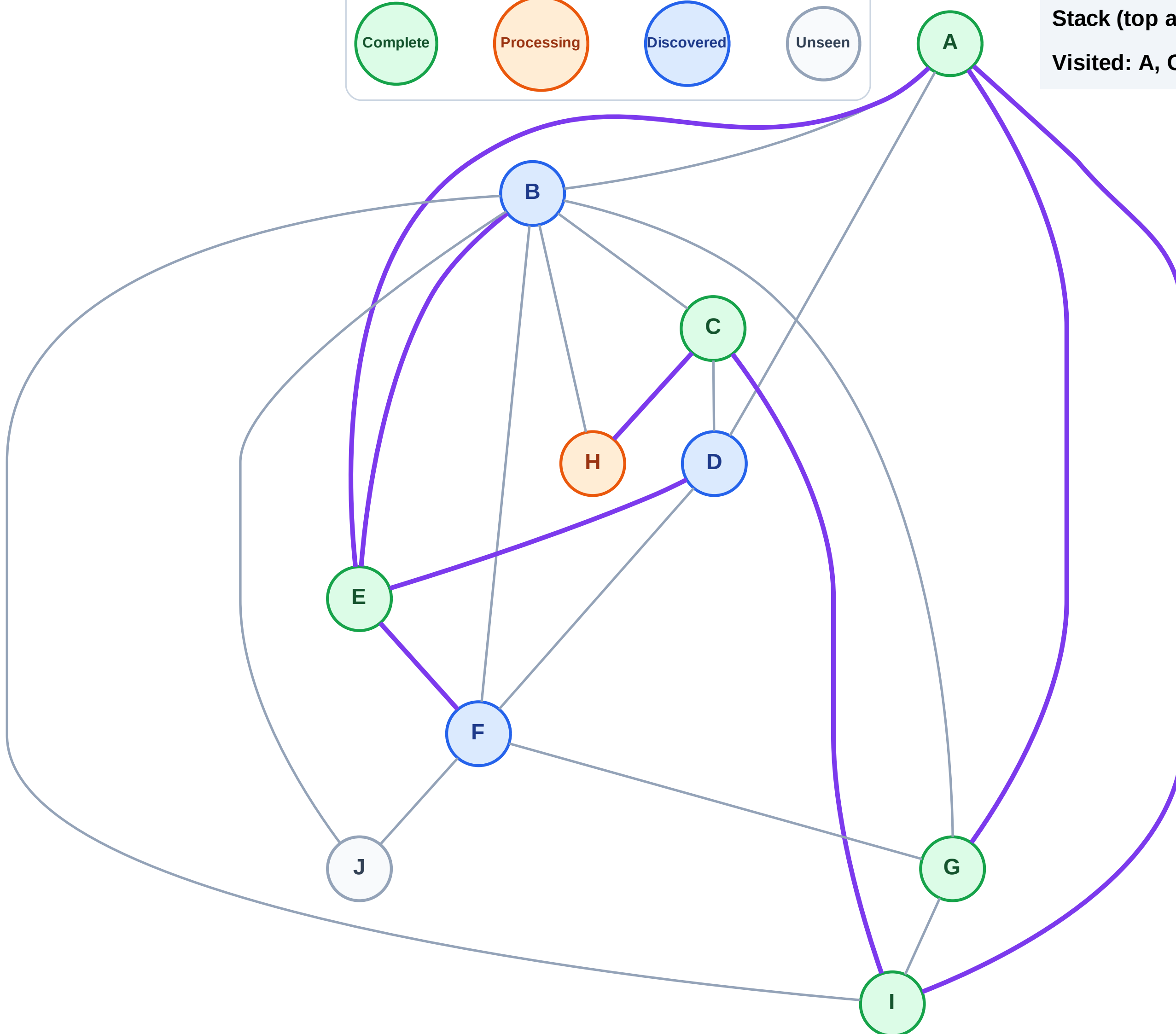
Step 12: Process node H

Legend



Stack (top at right): F | D | B

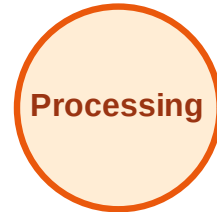
Visited: A, C, E, G, I



DFS Traversal

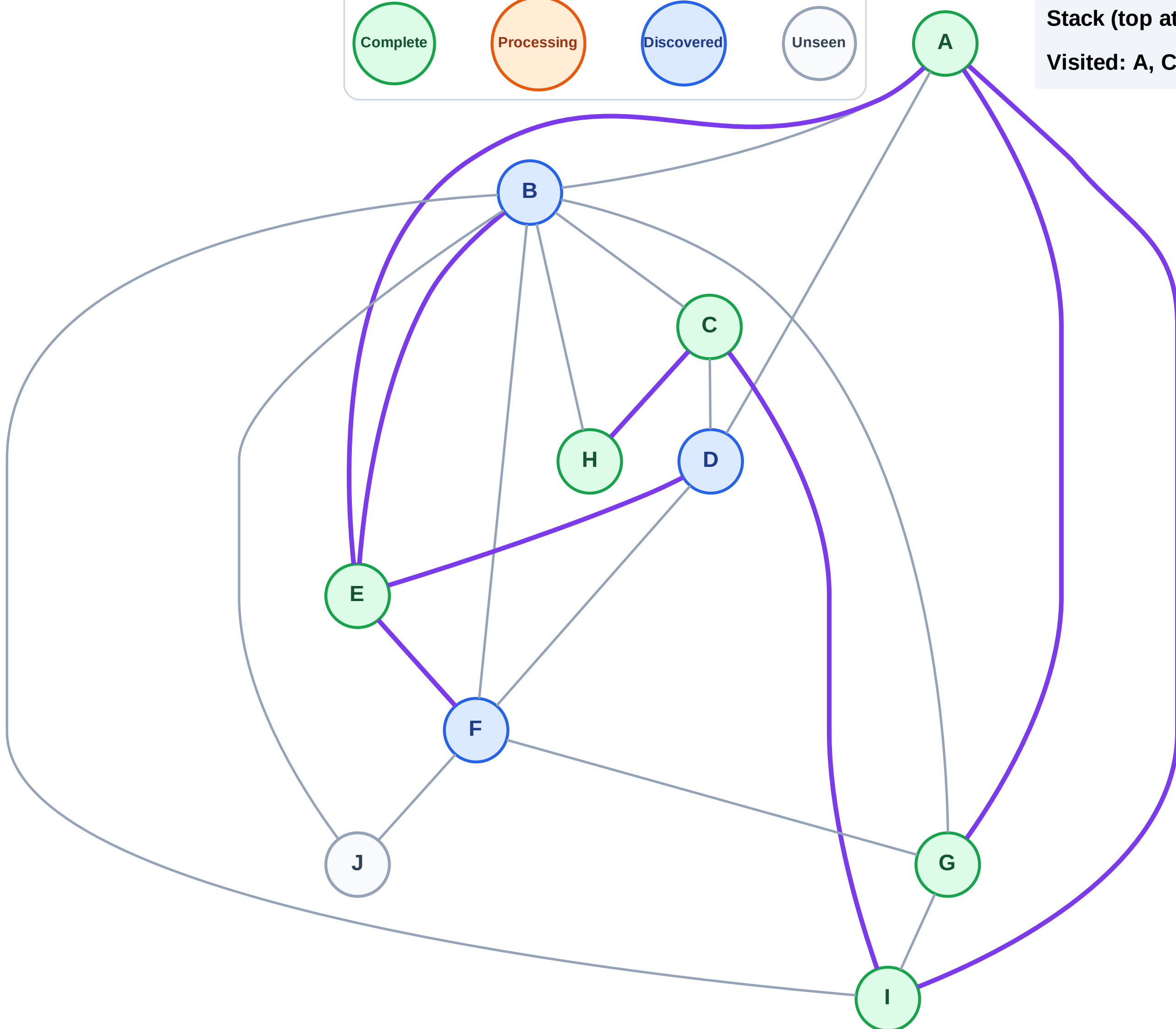
Step 13: No new neighbors from H

Legend



Stack (top at right): F | D | B

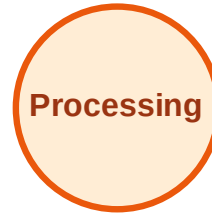
Visited: A, C, E, G, H, I



DFS Traversal

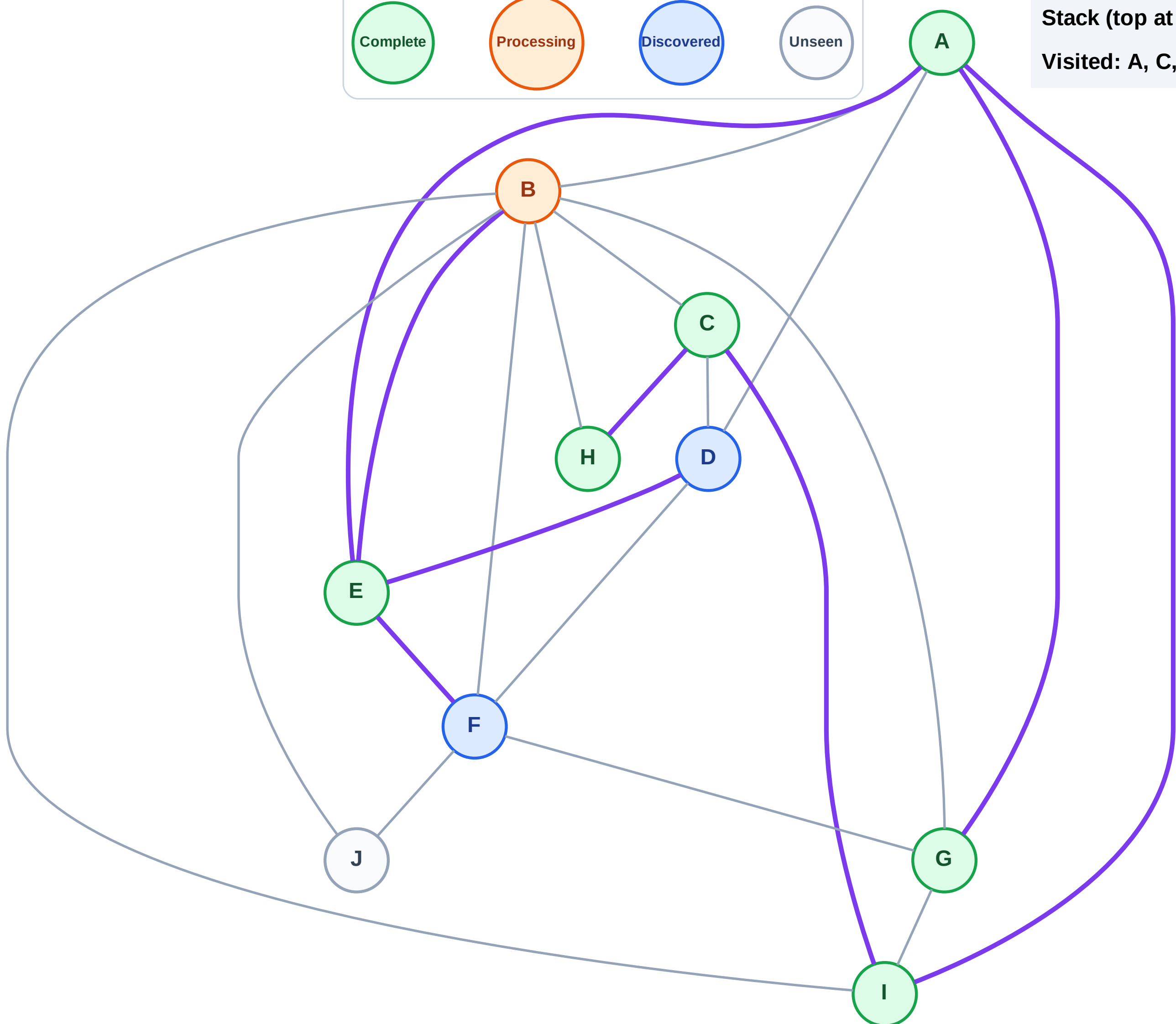
Step 14: Process node B

Legend



Stack (top at right): F | D

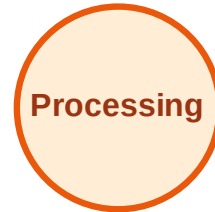
Visited: A, C, E, G, H, I



DFS Traversal

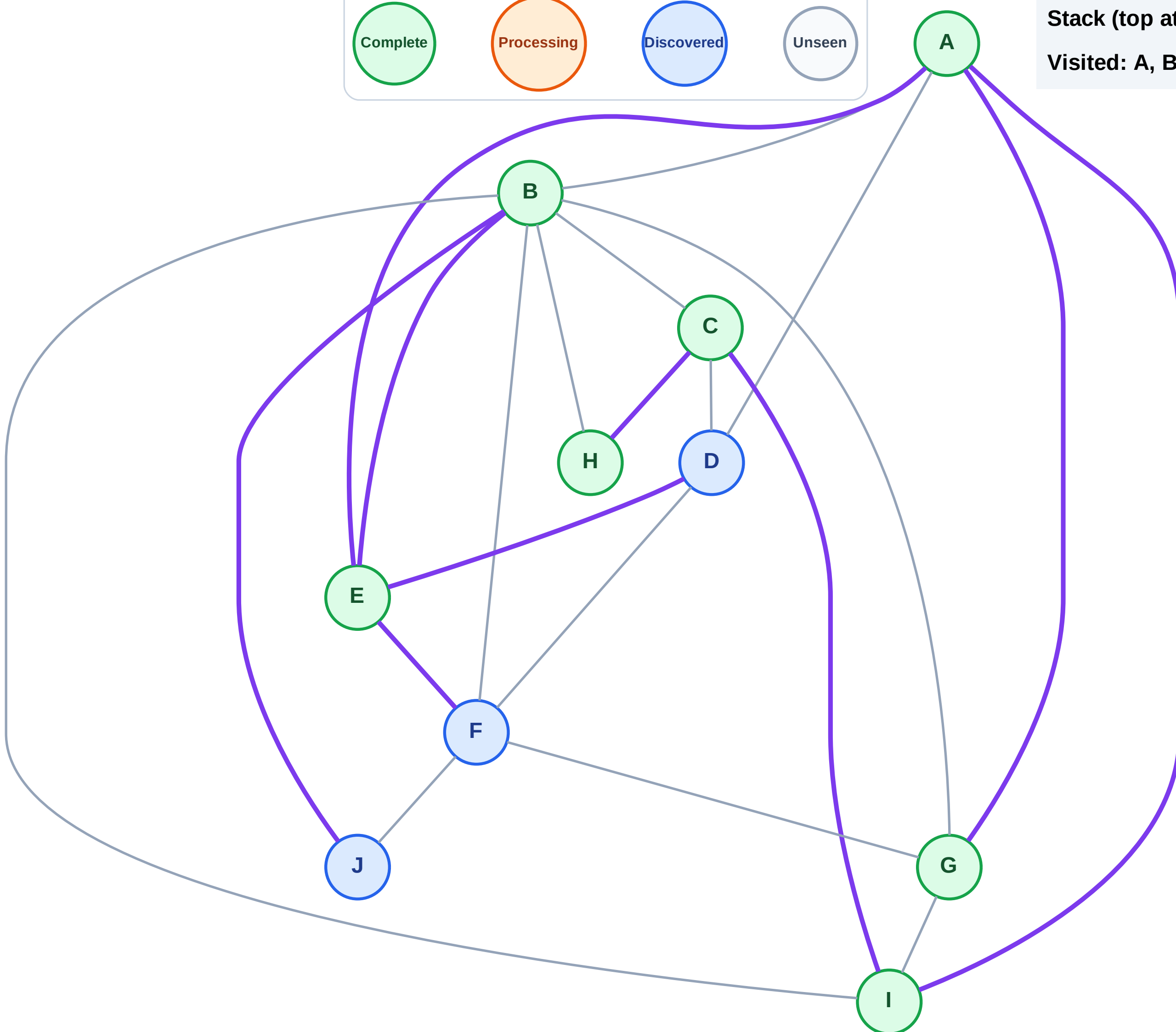
Step 15: Discover J from B

Legend



Stack (top at right): F | D | J

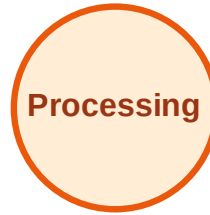
Visited: A, B, C, E, G, H, I



DFS Traversal

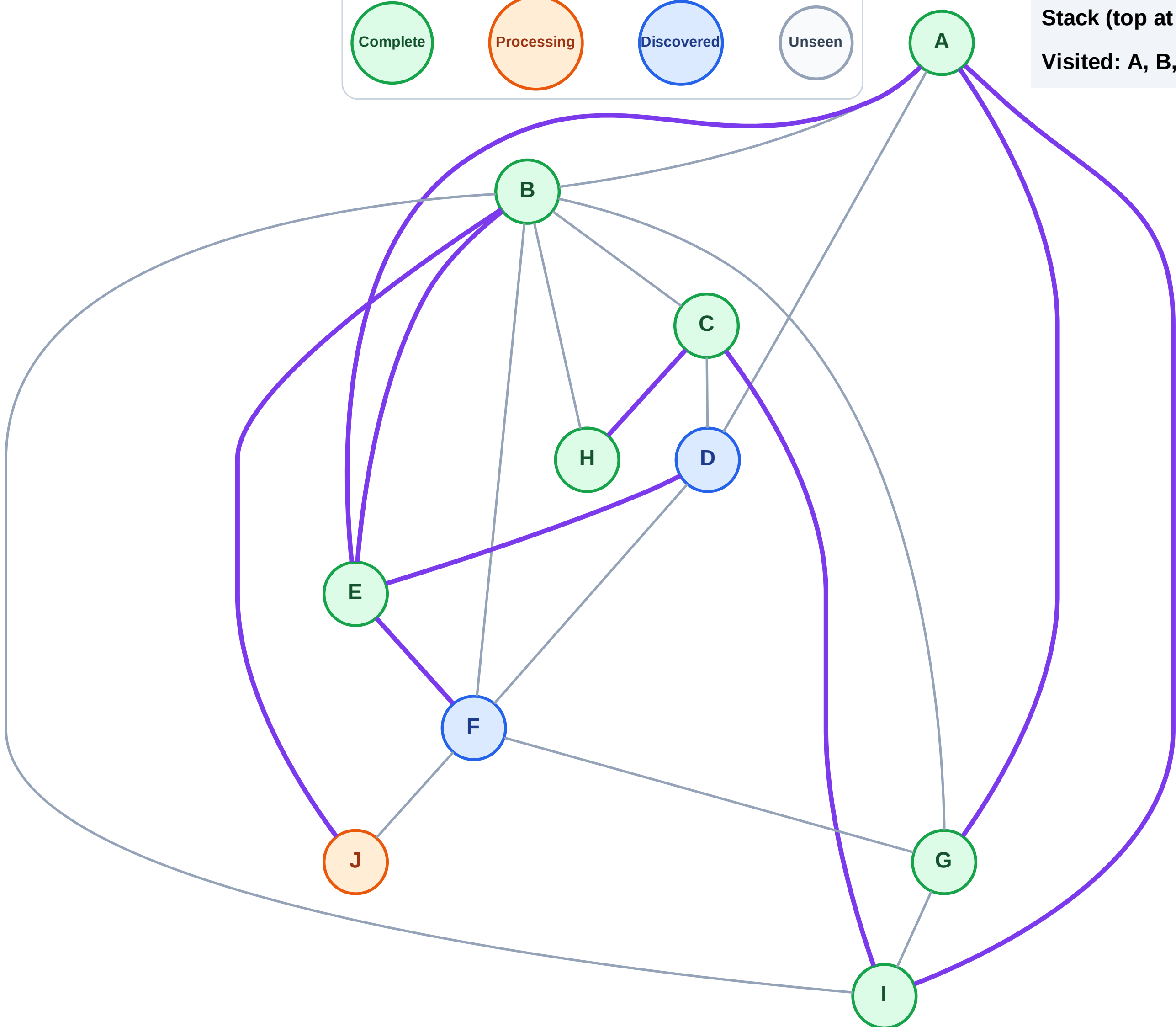
Step 16: Process node J

Legend



Stack (top at right): F | D

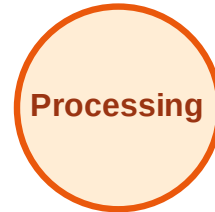
Visited: A, B, C, E, G, H, I



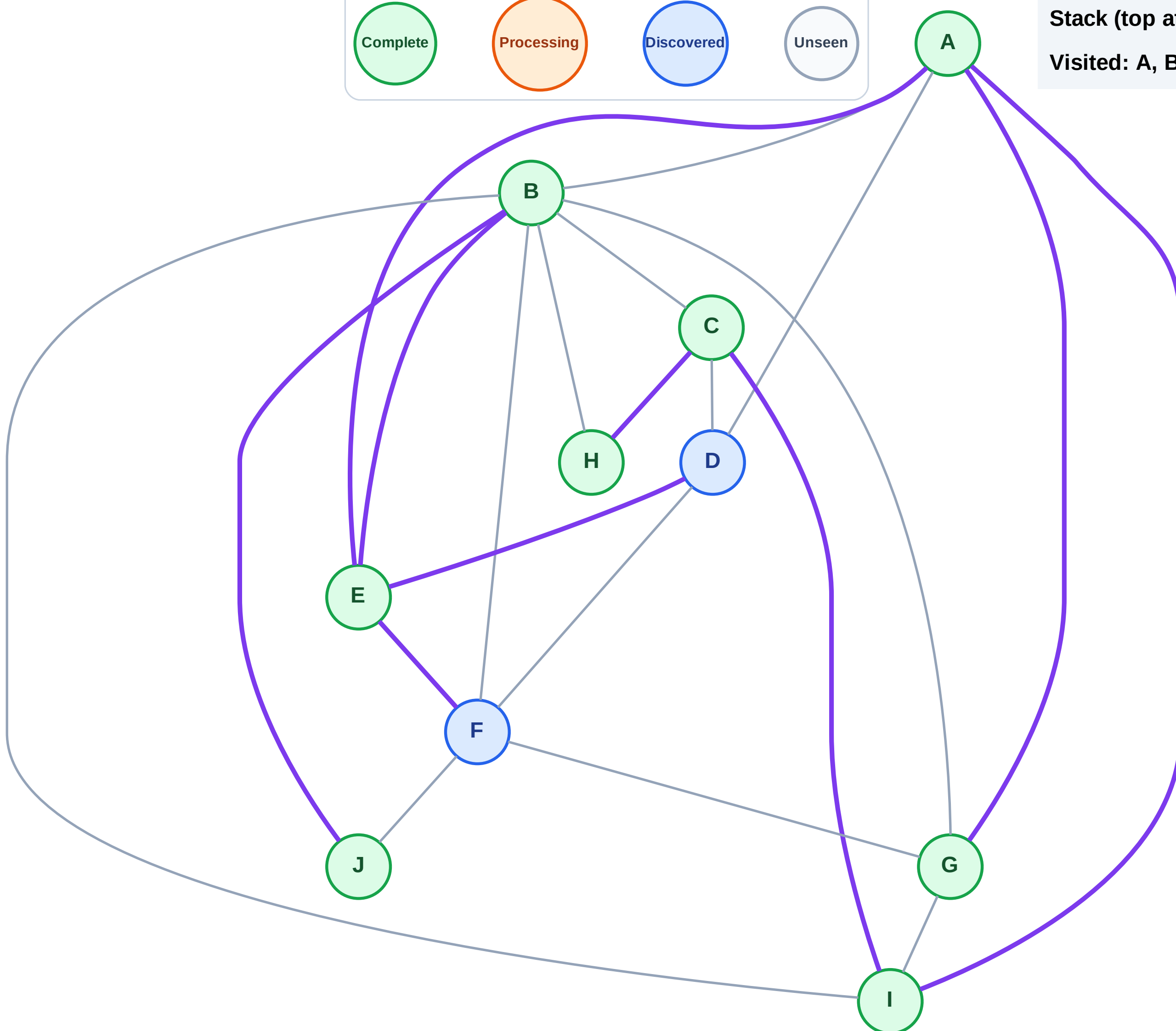
DFS Traversal

Step 17: No new neighbors from J

Legend



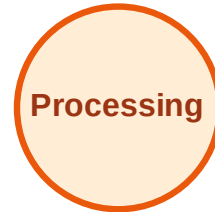
Stack (top at right): F | D
Visited: A, B, C, E, G, H, I, J



DFS Traversal

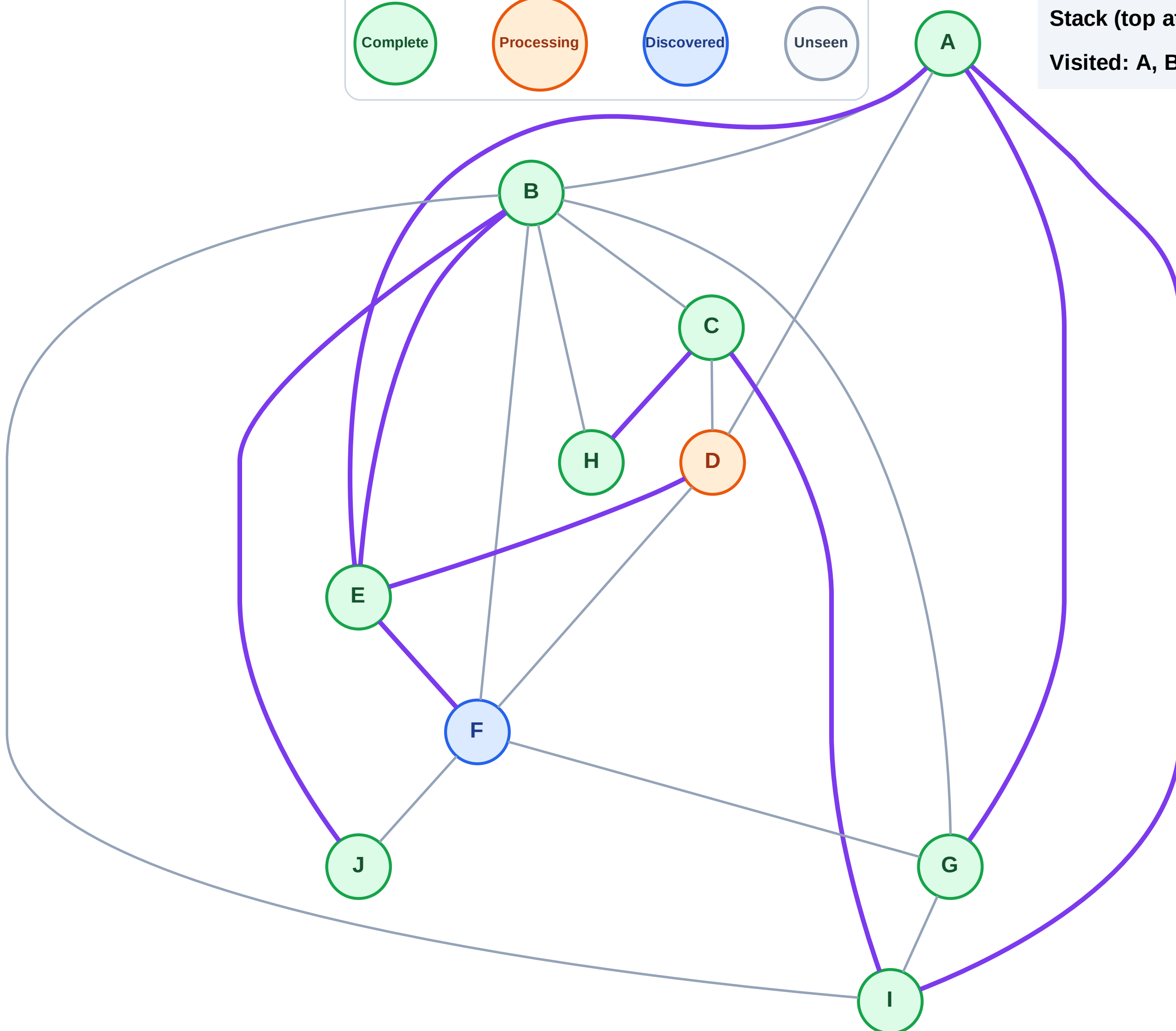
Step 18: Process node D

Legend



Stack (top at right): F

Visited: A, B, C, E, G, H, I, J



DFS Traversal

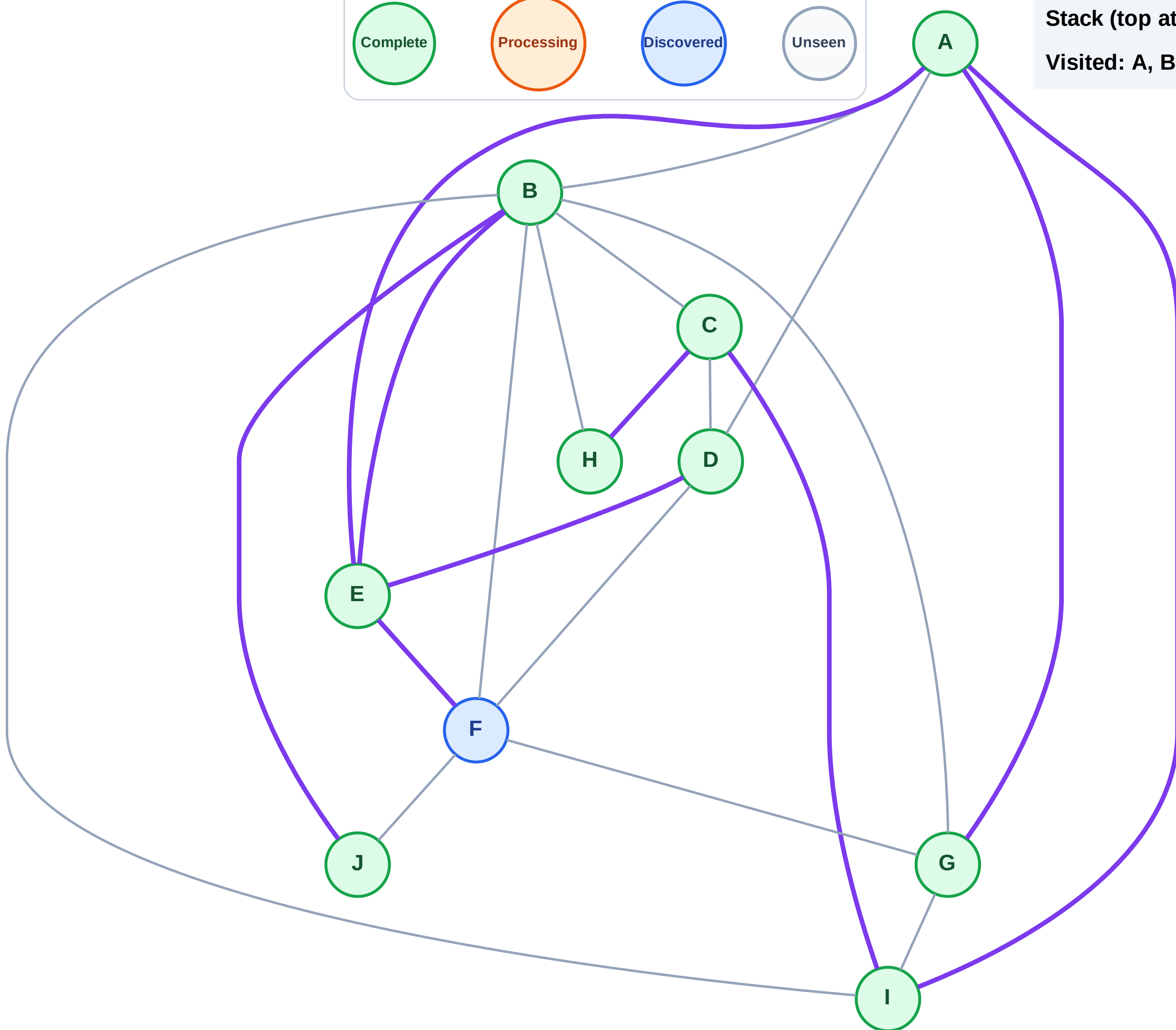
Step 19: No new neighbors from D

Legend



Stack (top at right): F

Visited: A, B, C, D, E, G, H, I, J



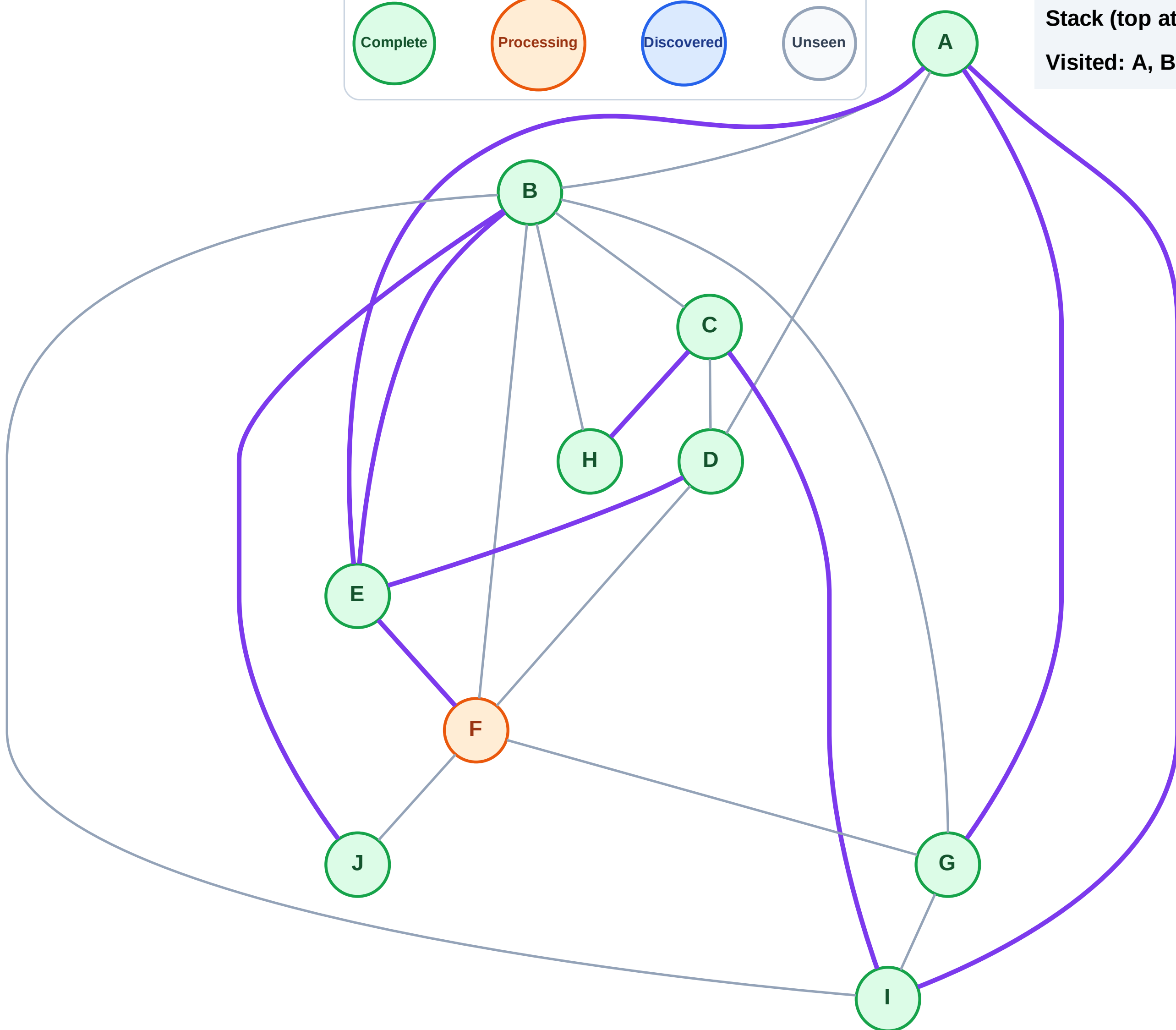
DFS Traversal

Step 20: Process node F

Legend



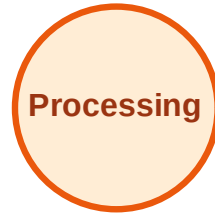
Stack (top at right): (empty)
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

